

Reading 5: Time Value of Money

1. Interest Rate (i)

- $i = R_f + \text{Inf P} + \text{Default Risk P} + \text{Liquidity P} + \text{Maturity P}$
- Nominal R_f i rate = Real R_f i Rate + Inf P
- i rate as a growth rate = $g = \left(\frac{FV_N}{PV}\right)^{\frac{1}{N}} - 1$

2. PV and FV of CF =

- $PV = \frac{FV}{(1+r)^N}$
- $PV \text{ of Perpetuity} = \frac{PMT}{r}$
- $PV \text{ (for more than one Compounding per year)} = PV = FV_N \left(1 + \frac{r_s}{m}\right)^{-m \times N}$
where $r_s = \text{stated ann } i - \text{rate}$
- $FV_N = PV(1+r)^N$
- $FV \text{ (for more than one Compounding per year)} = FV_N = \left(1 + \frac{r_s}{m}\right)^{m \times N}$
- $FV \text{ (for Continuous Compounding)} = FV_N = PV e^{r_s \times N}$
- Solving for $N = \frac{LN\left(\frac{FV}{PV}\right)}{LN(1+r)}$ (where LN = natural log)

4. Stated & Effective Rates

- Periodic i Rate = $\frac{\text{Stated Ann } i \text{ Rate}}{\text{No of Compounding Periods in One Year}}$
- Effective (or Equivalent) Ann Rate (EAR = EFF%) = $(1 + \text{Periodic } i \text{ Rate})^m - 1$

- EAR (with Continuous Compounding) = $\text{EAR} = e^{r_s} - 1$

5. PV & FV of Ordinary Annuity

- $PV_{OA} = \sum_{t=1}^n \frac{PMT}{(1+r)^t} = PMT \left[\frac{1 - \frac{1}{(1+r)^N}}{r} \right]$
- $FV_{OA} = \sum_{t=1}^n (PMT_t(1+r))^{N-t} = PMT \left[\frac{(1+r)^N - 1}{r} \right]$
- Size of Annuity Payment = $PMT = \frac{PV}{PV \text{ of Annuity Factor}}$
- $PV \text{ of Annuity Factor} = \frac{1 - \frac{1}{\left(1 + \left(\frac{r_s}{m}\right)\right)^{m \times N}}}{\frac{r_s}{m}}$

6. PV & FV of Annuity Due

- $PV_{AD} = PMT \left[\frac{1 - \frac{1}{(1+r)^N}}{r} \right] + PMT \text{ at } t = PV_{OA} + PMT$
- $FV_{AD} = PMT \left[\frac{(1+r)^N - 1}{r} \right] (1+r) = FV_{OA} \times (1+r)$

Reading 6: Discounted Cash Flow Applications

- $NPV = \sum_{t=1}^n \frac{CF_t}{(1+r)^t} - CF_0$
- IRR (when project's CFs are perpetuity) = $NPV = -I_0 + \frac{CF}{IRR} = 0$
- $HPR = \frac{(P_1 - P_0 + D_1)}{P_0}$

- $MMWR = \sum_{t=0}^T \frac{CF}{(1+IRR)^t} = 0$ (IRR represents the MWR)

5. TWR:

- TWR (when no external CF) = $r_{TWR} = HPR = r_t = \frac{MV_1 - MV_0}{MV_0}$
- TWR (for more than one periods) = $r_{TWR} = [(1+r_{t,1}) \times (1+r_{t,2}) \times \dots (1+r_{t,n})] - 1$
- Annualized TWR (when investment is for more than one year) = $[(1+R_1)(1+R_2 \dots + (1+R_n))]^{\frac{1}{n}} - 1$
- TWR (for the year) = $r_{TWR} = [(1+R_1) \times (1+R_2) \times \dots (1+R_{365})] - 1$ where $R_1 = \frac{MV_1 - MV_0}{MV_0}$

- Bank Discount Yield = $BDY = r_{BD} = \frac{360}{n} \frac{\text{Par} - \text{Price}}{\text{Par}}$ therefore $\text{Price} = \text{Par} \left(1 - \frac{n \times r_{BD}}{360}\right)$

- Holding Period Yield = $HPY = \frac{(P_1 - P_0 + D_1)}{P_0}$

- Effective Annual Yield = $EAY = (1 + HPY)^{365/t} - 1$ (Rule: $EAY > BDY$)

- Money Market Yield (or CD equivalent Yield) r_{MM} :

- $r_{MM} = HPY \times \left(\frac{360}{t}\right)$
- $r_{MM} = (r_{BD}) \times \frac{\text{Face Value of the Treasury Bill}}{\text{Purchase Price}}$

- $r_{MM} = \frac{360(r_{BD})}{360 - (t)(r_{BD})}$ (Rule: $r_{MM} > r_{BD}$)

10. Bond Equivalent Yield = BDY =
Semiannual Yield $\times 2$

Reading 7: Statistical Concepts & Market Returns

1. Range = Max Value – Min Value

2. Class Interval = $i \geq \frac{H-L}{k}$ where

- i = class interval
- H = highest value
- L = lowest value, k = No. of classes.

3. Absolute Frequency = Actual No of Observations (obvs) in a given class interval

4. Relative Frequency = $\frac{\text{Absolute Frequency}}{\text{Total No of Obvs}}$

5. Cumulative Absolute Frequency = Add up the Absolute Frequencies

6. Cumulative Relative Frequency = Add up the Relative Frequencies

7. Arithmetic Mean = $\frac{\text{Sum of obvs in database}}{\text{No. of obvs in the database}}$

8. Median = Middle No (when observations are arranged in ascending/descending order)

- For Even no of obvs locate median at $\frac{n}{2}$
- For Odd no. of obvs locate median at $\frac{n+1}{2}$

9. Mode = obvs that occurs most frequently in the distribution

10. Weighted Mean = $\bar{X}_w = \frac{\sum_{i=1}^n w_i X_i}{(w_1 X_1 + w_2 X_2 + \dots + w_n X_n)}$

11. Geometric Mean = $GM = \sqrt[n]{X_1 X_2 \dots X_n}$ with $X_i \geq 0$ for $i = 1, 2, \dots, n$.

12. Harmonic Mean = $H.M = \bar{X}_H = \frac{n}{\sum_{i=1}^n \left(\frac{1}{X_i}\right)}$

13. Population Mean = $\mu = \frac{\sum_{i=1}^n X_i}{N}$ with $X_i > 0$ for $i = 1, 2, \dots, n$.

14. Sample Mean = $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$ where n = number of observation in the sample

15. Measures of Location:

- Quartiles = $\frac{\text{Distribution}}{4}$
- Quintiles = $\frac{\text{Distribution}}{5}$
- Deciles = $\frac{\text{Distribution}}{10}$,
- Percentiles = $L_y = (n + 1) \frac{y}{100}$

16. Mean Absolute Deviation = $MAD = \frac{\sum_{i=1}^n |X_i - \bar{X}|}{n}$

17. Population Var = $\sigma^2 = \frac{\sum_{i=1}^N (X_i - \mu)^2}{N}$

18. Population S.D = $\sqrt{\sigma^2} = \sqrt{\frac{\sum_{i=1}^N (X_i - \mu)^2}{N}}$

19. Sample Var = $s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$

20. Sample S.D = $s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$

21. Semi-var = $\sum_{\text{For all } X_i \leq \bar{X}} \frac{(X_i - \bar{X})^2}{n-1}$

22. Semi-deviation (Semi S.D) =

$$\sqrt{\text{semivariance}} = \sqrt{\sum_{\text{For all } X_i \leq \bar{X}} \frac{(X_i - \bar{X})^2}{n-1}}$$

23. Target Semi-var = $\sum_{\text{For all } X_i \leq B} \frac{(X_i - B)^2}{n-1}$ where B = Target Value

24. Target Semi-Deviation = $\sqrt{\text{target semivariance}} = \sqrt{\sum_{\text{For all } X_i \leq B} \frac{(X_i - B)^2}{n-1}}$

25. Coefficient of Variation = $CV = \left(\frac{s}{\bar{X}}\right)$ where s = sample S.D and $\bar{X}$ = sample mean

26. Sharpe Ratio = $\frac{\text{Mean Portfolio R} - \text{Mean Rf R}}{\text{S.D of Portfolio R}}$

27. Excess Kurtosis = Kurtosis – 3

28. Geometric Mean $R \approx$

$$\text{Arithmetic Mean } R - \frac{\text{Variance of } R}{2}$$

Reading 8: Probability Concepts

1. Empirical Prob of an event $E = P(E) =$

$$\frac{\text{Prob of event } E}{\text{Total Prob}}$$

2. Odds for event $E = \frac{\text{Prob of } E}{1 - \text{Prob of } E}$

3. Odds against event $E = \frac{1 - \text{Prob of } E}{\text{Prob of } E}$

4. Conditional Prob of A given that B has occurred $= P(A|B) = \frac{P(AB)}{P(B)} \rightarrow P(B) \neq 0$.

5. Multiplication Rule (Joint probability that both events will happen):

$$P(A \text{ and } B) = P(AB) = P(A|B) \times P(B)$$

$$P(B \text{ and } A) = P(BA) = P(B|A) \times P(A)$$

6. Addition Rule (Prob that event A or B will occur):

$$P(A \text{ or } B) = P(A) + P(B) - P(AB)$$

$$P(A \text{ or } B) = P(A) + P(B) \quad (\text{when events are mutually exclusive because } P(AB) = 0)$$

7. Independent Events:

- Two events are independent if:
 $P(B|A) = P(B)$ or if $P(A|B) = P(A)$

- Multiplication Rule for two independent events $= P(A \& B) = P(AB) = P(A) \times P(B)$
- Multiplication Rule for three independent events $= P(A \text{ and } B \text{ and } C) = P(ABC) = P(A) \times P(B) \times P(C)$

8. Complement Rule (for an event $S) = P(S) + P(S^C) = 1$ (where S^C is the event not S)

9. Total Probability Rule:
 $P(A) = P(AS) + P(AS^C) = P(A|S) \times P(S) + P(A|S^C) \times P(S^C)$
 $P(A) = P(AS_1) + P(AS_2) + \dots + P(AS_n) = P(A|S_1) \times P(S_1) + P(A|S_2) \times P(S_2) \dots P(A|S_n) \times P(S_n)$

(where $S_1, S_2, \dots, S_n$ are mutually exclusive and exhaustive scenarios)

10. Expected $R = E(w_i R_i) = w_i E(R_i)$

11. $\text{Cov}(R_i, R_j) = \sum_{i=1}^n (p(R_i - ER_i))(R_j - ER_j)$
 $\text{Cov}(R_i, R_j) = \text{Cov}(R_j, R_i)$
 $\text{Cov}(R, R) = \sigma^2(R)$

12. Portfolio $\text{Var} = \sigma^2(R_p) =$

$$\sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(R_i, R_j)$$

$$\sigma^2(R_p) = w_1^2 \sigma^2(R_1) + w_2^2 \sigma^2(R_2) + w_3^2 \sigma^2(R_3) + 2w_1 w_2 \text{Cov}(R_1, R_2) + 2w_1 w_3 \text{Cov}(R_1, R_3) + 2w_2 w_3 \text{Cov}(R_2, R_3)$$

13. Standard Deviation (S.D) =

$$\sqrt{w_1^2 R_i + w_2^2 R_2 + w_3^2 R_3}$$

14. Correlation (b/w two random variables $R_i, R_j) = \rho(R_i, R_j) = \frac{\text{Cov}(R_i, R_j)}{\sigma_{R_i} \times \sigma_{R_j}}$

15. Bayes' Formula =

$$\frac{P(\text{Event}|\text{New Information})}{\frac{P(\text{New Information}|\text{Event})}{P(\text{New Information})} \times P(\text{Prior prob. of Event})}$$

16. Multiplication Rule of Counting = n factorial $= n! = n(n-1)(n-2)(n-3) \dots 1$.

17. Multinomial Formula (General formula for labeling problem) $= \frac{n!}{n_1! n_2! \dots n_k!}$

18. Combination Formula (Binomial Formula)

$$= {}^n C_r = \binom{n}{r} = \frac{n!}{(n-r)! r!}$$

where n = total no. of objects and r = no. of objects selected.

19. Permutation $= {}^n P_r = \frac{n!}{(n-r)!}$

Reading 9: Common Probability Distributions

1. Probability Function (for a binomial random variable) $p(x) = P(X=x) = \binom{n}{x} p^x (1-p)^{n-x} = \frac{n!}{(n-x)! x! p^x} (1-p)^{n-x}$ (for $x = 0, 1, 2, \dots, n$)

- x = success out of n trials
 - $n-x$ = failures out of n trials
 - p = probability of success
 - $1-p$ = probability of failure
 - n = no of trials.
2. Probability Density Function (pdf) = $f(x)$

$$= \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$F(x) = \frac{x-a}{b-a} \text{ for } a < x < b$$
3. Normal Density Funct = $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ for $-\infty < x < +\infty$
4. Estimations by using Normal Distribution:
- Approximately 50% of all obsv fall in the interval $\mu \pm \frac{2}{3}\sigma$
 - Approx 68% of all obsvs fall in the interval $\mu \pm \sigma$
 - Approx 95% of all obsvs fall in the interval $\mu \pm 2\sigma$
 - Approx 99% of all obsvs fall in the interval $\mu \pm 3\sigma$
 - More precise intervals for 95% of the obsvs are $\mu \pm 1.96\sigma$ and for 99% of the observations are $\mu \pm 2.58\sigma$.
5. Z-Score (how many S.Ds away from the mean the point x lies) $z = \frac{x-\mu}{\sigma}$ (when X is normally distributed)
6. Roy's Safety-Frist Criterion = SF Ratio = $\frac{[E(R_P) - R_L]}{\sigma_P}$
7. Sharpe Ratio = $\frac{[E(R_P) - R_f]}{\sigma_P}$
8. Value at Risk = VAR = Minimum \$ loss expected over a specified period at a specified prob level.
9. Mean (μ_L) of a lognormal random variable = $\exp(\mu + 0.50\sigma^2)$
10. Variance (σ_L^2) of a lognormal random variable = $\exp(2\mu + \sigma^2) \times [\exp(\sigma^2) - 1]$.
11. Log Normal Price = $S_T = S_0 \exp(r_{0,T})$
 Where, $\exp = e$ and $r_{0,t}$ = Continuously compounded return from 0 to T
12. Price relative = End price / Beg price = $S_{t+1} / S_t = 1 + R_{t,t+1}$
where,
 $R_{t,t+1}$ = holding period return on the stock from t to $t+1$.
13. Continuously compounded return associated with a holding period from t to $t+1$:
 $r_{t,t+1} = \ln(1 + \text{holding period return})$ or
 $r_{t,t+1} = \ln(\text{price relative}) = \ln(S_{t+1} / S_t) = \ln(1 + R_{t,t+1})$
14. Continuously compounded return associated with a holding period from 0 to T :
 $R_{0,T} = \ln(S_T / S_0)$ or $r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \dots + r_{0,1}$
 Where,
 $r_{T-1,T}$ = One-period continuously compounded returns
15. When one-period continuously compounded returns (i.e. $r_{0,1}$) are IID random variables.

$$E(r_{0,T}) = E(r_{T-1,T}) + E(r_{T-2,T-1}) + \dots + E(r_{0,1}) = \mu T$$
 And

$$\text{Variance} = \sigma^2(r_{0,T}) = \sigma^2 T$$

$$\text{S.D.} = \sigma(r_{0,T}) = \sigma\sqrt{T}$$
16. Annualized volatility = sample S.D. of one period continuously compounded returns $\times \sqrt{T}$

Reading 10: Sampling and Estimation

1. Var of the distribution of the sample mean

$$= \frac{\sigma^2}{n}$$
2. S.D of the distribution of the sample mean

$$= \sqrt{\frac{\sigma^2}{n}}$$

3. Standard Error of the sample mean:
 - When the population S.D (σ) is known
 $= \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$
 - When the population S.D (σ) is not known
 $= s_{\bar{X}} = \frac{s}{\sqrt{n}}$ where s = sample S.D estimate of $s = \frac{\sqrt{\text{sample variance}}}{\sqrt{s^2}}$ where $s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$
4. Finite Population Correction Factor = fpc
 $= \sqrt{\frac{N-n}{N-1}}$ where N = population
5. New Adjusted Estimate of Standard Error
 $= (\text{Old estimated standard error} \times \text{fpc})$
6. Construction of Confidence Interval (CI) =
 Point estimate $\pm$ (Reliability factor $\times$ Standard error)
 - CI for normally distributed population with known variance $= \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
 - CI for normally distributed population with unknown variance $= \bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$ where S = sample S.D.
7. Student's t distribution
 $\mu = \bar{X} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$
8. Z-ratio $= Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

$$9. \text{ t-ratio} = t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Reading 11: Hypothesis Testing

1. Test Statistic =

$$\frac{\text{Sample Statistic} - \text{Hypothesized Value of pop parameter}}{\text{standard error of sample statistic}^*}$$
 - * when Pop S.D is unknown, the standard error of sample statistic is give by $S_{\bar{X}} = \frac{s}{\sqrt{n}}$
 - * when Pop S.D is unknown, the standard error of sample statistic is give by $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$
2. Power of Test = 1-Prob of Type II Error
3. $z = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$ (when sample size is large or small but pop S.D is known)
4. $z = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$ (when sample size is large but pop S.D is unknown where s is sample S.D)
5. $t_{n-1} = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$ (when sample size is large or small and pop S.D is unknown and pop sampled is normally or approximately normally distributed)

6. Test Statistic for a test of diff b/w two pop means (normally distributed, pop var unknown but assumed equal)

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\left(\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2} \right)^{1/2}} \text{ where } S_p^2 = \text{pooled}$$

estimator of common variance =

$$\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \text{ where } df = n_1 + n_2 - 2.$$

7. Test Statistic for a test of diff b/wn two pop means (normally distributed, unequal and unknown pop var unknown)

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^{1/2}} \text{ In this df calculated as}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2}}$$

8. Test Statistic for a test of mean differences (normally distributed populations, unknown population variances)

$$\bullet \quad t = \frac{\bar{d} - \mu_{d0}}{s_{\bar{d}}}$$

$$\bullet \quad \text{sample mean difference} = \bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$$

$$\bullet \quad \text{sample variance} = S_d^2 = \frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n-1}$$

$$\bullet \quad \text{sample S.D} = \sqrt{S_d^2}$$

- sample error of the sample mean difference = $s \bar{d} = \frac{s_d}{\sqrt{n}}$
8. Chi Square Test Statistic (for test concerning the value of a normal population variance) $X^2 = \frac{(n-1)S^2}{\sigma_0^2}$ where $(n-1) = df$ and $S^2 = \text{sample variance} = \frac{\sum_{i=0}^n (X_i - \bar{X})^2}{n-1}$
9. Chi Square Confidence Interval for variance
Lower limit = $L = \frac{(n-1)S^2}{\chi^2_{a/2}}$ and Upper limit = $U = \frac{(n-1)S^2}{\chi^2_{1-a/2}}$
10. F-test (test concerning differences between variances of two normally distributed populations) $F = \frac{S_1^2}{S_2^2}$

 $S_1^2 = 1\text{st sample var with } n_1 \text{ obs}$
 $S_2^2 = 2\text{nd sample var with } n_2 \text{ obs}$
 $df_1 = n_1 - 1 \text{ numerator df}$
 $df_2 = n_2 - 1 \text{ denominator df}$
11. Relation between Chi Square and F-distribution = $F = \frac{\chi^2_1/m}{\chi^2_2/n}$ where:
- χ^2_1 is one chi square random variable with one m degrees of freedom

- χ^2_2 is another chi square random variable with one n degrees of freedom
12. Spearman Rank Correlation = r_s
$$= 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$
- For small samples rejection points for the test based on r_s are found using table.
 - For large sample size (e.g. $n > 30$) t-test can be used to test the hypothesis i.e.
$$t = \frac{(n-2)^{1/2} r_s}{(1 - r_s^2)^{1/2}}$$

Reading 12: Technical Analysis

1. Relative Strength Analysis = $\frac{\text{Price of asset}}{\text{Price of the Benchmark Asset}}$
2. Price Target for the
- Head and Shoulders = Neckline – (Head – Neckline)
 - Inverse Head and Shoulders = Neckline + (Neckline – Head)
3. Simple Moving Average = $\frac{P_1 + P_2 + P_3 \dots + P_n}{N}$
4. Momentum Oscillator (or Rate of Change Oscillator ROC):
- Momentum Oscillator Value M = $(V - V_x) \times 100$

- (where V = most recent closing price and V_x = closing price x days ago)
- Alternate Method to calculate M = $\frac{V}{V_x} \times 100$
5. Relative Strength Index = $RSI = 100 - \frac{100}{1 + RS}$ where
$$RS = \frac{\sum(\text{Up changes})}{\sum(\text{Down changes})}$$
6. Stochastic Oscillator (composed of two lines %K and %D):
- $\%K = 100 \left(\frac{C - L14}{H14 - L14} \right)$ where:
C = latest closing price, L14 = lowest price in last 14 days, H14 is highest price in last 14 days
 - $\%D$ = Average of the last three %K values calculated daily.
7. Put/Call Ratio (Type of Sentiment Indicators) = $\frac{\text{Volume of Put Options Traded}}{\text{Volume of Call Options Traded}}$
8. Short Interest Ratio (Type of Sentiment Indicators) = $\frac{\text{Short Interest}}{\text{Average Daily Trading Volume}}$
9. Arms Index TRIN i.e. Trading Index (Type of Flow of funds Indicator) =
$$\text{Arm Index or TRIN} = \frac{\text{No. of Advan Issues} \div \text{No. of Declin Issues}}{\text{Volume of Advan Issues} \div \text{Volume of Declin Issues}}$$

Reading 13: Demand & Supply Analysis: Introduction

1. Slope of the demand curve =
$$\frac{\Delta \text{ in Price}}{\Delta \text{ in Quantity Demanded}}$$
2. Slope of the supply curve =
$$\frac{\Delta \text{ in Price}}{\Delta \text{ in Quantity Supplied}}$$
3. Consumer Surplus = Value that a consumer places on units consumed – Price paid to buy those units
 - Area (for calculating Consumer Surplus) = $\frac{1}{2} (\text{Base} \times \text{Height}) = \frac{1}{2} (Q_0 \times P_0)$
4. Producer Surplus = Total revenue received from selling a given amount of a good – Total variable cost of producing that amount
 - Total revenue = Total quantity sold $\times$ Price per unit
 - Area (for calculating Producer Surplus) = $\frac{1}{2} (\text{Base} \times \text{Height}) = \frac{1}{2} \{(Q_0) \times (P_0 - \text{intercept point on y-axis}^{**})\}$

***where supply curve intersects y-axis*
5. Total Surplus = Consumer surplus + Producer surplus

6. Total Surplus = Total value – Total variable cost
7. Society Welfare = Consumer surplus + Producer surplus
8. Price Elasticity of Demand =
$$\frac{\% \Delta \text{ in Quantity Demanded}}{\% \Delta \text{ in Price}}$$

$$\frac{\% \Delta Q}{\% \Delta P} = \frac{\frac{Q_2 - Q_1}{\frac{1}{2}(Q_1 + Q_2)}}{\frac{P_2 - P_1}{\frac{1}{2}(P_1 + P_2)}}$$
9. Income Elasticity of Demand =
$$\frac{\% \Delta \text{ in Quantity Demanded}}{\% \Delta \text{ in Income}}$$

$$\frac{\% \Delta Q}{\% \Delta I} = \frac{\frac{Q_2 - Q_1}{\frac{1}{2}(Q_1 + Q_2)}}{\frac{I_2 - I_1}{\frac{1}{2}(I_1 + I_2)}}$$
10. Cross Elasticity =
$$\frac{\% \Delta \text{ in Quantity Demanded of Good X}}{\% \Delta \text{ in Price of Good Y}}$$

Reading 14: Demand & Supply Analysis: Consumer Demand

1. Marginal Utility =
$$\frac{\Delta \text{ in Total Utility}}{\Delta \text{ in Quantity Consumed}}$$
2. Equation of Budget Constraint Line = $(P_X \times Q_X) + (P_Y \times Q_Y)$

3. Slope of Budget Constraint Line =
$$\frac{\frac{\Delta \text{ in } Q_Y}{\Delta \text{ in } Q_X}}{\frac{P_X}{P_Y}}$$
4. Marginal Rate of Substitution =
$$\frac{\frac{\Delta \text{ in } Q_Y}{\Delta \text{ in } Q_X}}{\frac{\text{Marginal Utility of Good X}}{\text{Marginal Utility of Good Y}}}$$

Reading 15: Demand & Supply Analysis: The Firm

1. Profit = Total revenue – Total cost
2. Accounting Profit = Total Revenue – Explicit Costs (or Accounting costs)
3. Economic Profit
 - = Total Revenue – Explicit Costs – Implicit Costs or
 - = Accounting Profit – Implicit Costs or
 - = Total Revenue – Total Economic Costs
4. Economic costs = Explicit costs + Implicit costs
5. Normal Profit = Accounting Profit – Economic Profit
6. Accounting profit = Economic Profit + Normal Profit
7. Economic rent = (New “Higher” Price after $\uparrow$ in Demand – Previous Price before $\uparrow$ in Demand) $\times$ Q_S before $\uparrow$ in Demand

8. Total Revenue (TR):
 - = Price $\times$ Quantity or
 - = Sum of individual units sold $\times$ Respective prices of individual Units sold = $\sum (P_i \times Q_i)$
9. Average Revenue (AR) = $\frac{\text{Total Revenue}}{\text{Quantity}}$
10. Marginal Revenue (MR) = $\frac{\Delta \text{ in Total Revenue}}{\Delta \text{ in Quantity}}$
11. Total Variable Cost = Variable Cost per unit $\times$ Quantity Produced
12. Total Cost = Total Fixed + Total Variable
13. Average total cost (ATC) = $\frac{\text{Total Cost}}{\text{Quantity Produced}} = \text{Avg. Fixed Cost} + \text{Avg. Variable Cost}$
14. Marginal cost (MC) = $\frac{\Delta \text{ in Total Cost}}{\Delta \text{ in Quantity Produced}}$
15. Marginal Variable Cost = $\frac{\Delta \text{ in Total Variable Cost}}{\Delta \text{ in Quantity Produced}}$
16. Marginal revenue (in perfect competition) = Avg. Revenue = Price = Demand
17. Profit can be increased by increasing output when $MR > MC$
18. Profit can be increased by decreasing output when $MR < MC$
19. Break-even price: $P = ATC \rightarrow$ Output level where Price = Average Revenue = Marginal Revenue = Average Total Cost $\rightarrow$ where, Total Revenue = Total Cost.
20. Firms earn Economic Profits when Price $>$ Average Total Cost
21. Profits occur when Total Revenue (TR) $\geq$ Total Cost (TC) & when Price = Marginal Cost $\rightarrow$ firm will continue operating.
22. Losses are incurred when there are Operating profits (Total Revenue $\geq$ Variable Cost) but Total Revenue $<$ Total Fixed Cost + Total Variable Cost AND when Price = Marginal Cost while losses are $<$ fixed costs $\rightarrow$ firm will continue operating.
23. Losses are incurred when there are Operating losses (Total Revenue $\leq$ Variable Cost) AND when losses $\geq$ fixed costs $\rightarrow$ firm will shut down.
24. Average Product = $\frac{\text{Total Product}}{\text{Quantity of Labor}}$
25. Marginal Product = $\frac{\Delta \text{ in Total Product}}{\Delta \text{ in Quantity of Labor}} = \frac{\Delta \text{ in Total Output}}{\Delta \text{ in No of Workers}}$
26. Least-cost optimization Rule: $\frac{\text{Marginal Product of Labor}}{\text{Price of Labor}} = \frac{\text{Marginal Product of Physical Capital}}{\text{Price of Physical Capital}}$
27. Profit is maximized when: $MRP =$ Price or cost of the input for each type of resource that is used in the production process
28. Marginal Revenue product = Marginal Product of an input unit $\times$ Price of the Product = Price of the input = $\frac{\Delta \text{ in Total Revenue}}{\Delta \text{ in Quantity of Input employed}}$
29. Surplus value or contribution of an input to firm's profit = $MRP - \text{Cost of an input}$

Reading 16: The firm & Market Structures

1. In perfect competition, Marginal revenue = Avg. Revenue = Price = Demand
2. Marginal Revenue = Price $\times \left(1 - \frac{1}{\text{Price Elasticity of Demand}}\right)$
3. Concentration Ratio = $\frac{\text{Sum of sales values of the largest 10 firms}}{\text{Total Market Sales}}$
4. Herfindahl-Hirshman Index = Sum of the squares of the market shares of the top N companies in an industry

Reading 17: Aggregate Output, Prices & Economic Growth

1. Nominal GDP_t = Prices in year t × Quantity produced in year t
2. Real GDP_t = Prices in the base year × Quantity produced in year t
3. Implicit price deflator for GDP or GDP deflator =
$$\frac{\text{value of current yr output at current yr prices}}{\text{value of current yr output at base yr prices}} \times 100$$
4. Real GDP = [(Nominal GDP / GDP deflator) ÷ 100]
5. GDP deflator =
$$\frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$
6. GDP = Consumer spending on final good & services + Gross private domestic invst + Govt. spending on final goods & services + Govt. gross fixed invst + Exp – Imp + Statistical discrepancy
7. Net Taxes = Taxes – Transfer payments
8. GDP = National income + Capital consumption allowance + Statistical discrepancy
9. National Income = Compensation of employees + Corp & Govt enterprise profits before taxes + Interest income + unincorporated business net income + rent + indirect business taxes less subsidies
10. Total Amount Earned by Capital = Profit + Capital Consumption Allowance
11. PI = National income – Indirect business taxes – Corp income taxes – Undistributed Corp profits + Transfer payments
12. Personal disposable income (PDI) = Personal income – Personal taxes OR GDP (Y) + Transfer payments (F) – (R/E + Depreciation) – direct and indirect taxes (R)
13. Business Saving = R/E + Depreciation
14. Household saving = PDI - Consumption expenditures - Interest paid by consumers to business - Personal transfer payments to foreigners
15. Business sector saving = Undistributed corporate profits + Capital consumption allowance
16. Total Expenditure = Household consumption (C) + Investments (I) + Government spending (G) + Net exports (X-M)
17. Private Sector Saving = Household Saving + Undistributed Corporate Profits + Capital Consumption Allowance
18. GDP = Household consumption + Private Sector Saving + Net Taxes
19. Domestic saving = Investment + Fiscal balance + Trade balance
20. Trade Balance = Exports – Imports
21. Fiscal balance = Government Expenditure – Taxes = (Savings – Investment) – Trade Balance
22. Average propensity to consume (APC) =
$$\frac{\text{Aggregate Consumption}}{\text{Real Income}}$$
23. Quantity theory of money equation: Nominal Money Supply × Velocity of Money = Price Level × Real Income or Expenditure
24. % Δ in unit labor cost = % Δ in nominal wages - % Δ in productivity
25. Economic growth = Annual % Δ in real GDP
26. Total Factor Productivity growth = Growth in potential GDP – [Relative share of labor in National Income × (Growth in labor) + [Relative share of capital in National Income × (Growth in capital)]]
27. Growth in potential GDP = Growth in technology + (Relative share of labor in National Income × Growth in Labor) +

(Relative share of capital in National Income $\times$ Growth in capital]

28. Capital share = Corporate profits + net interest income + net rental income + (depreciation/ GDP)

29. Labor share = $\frac{\text{Employee Compensation}}{\text{GDP}}$

Reading 18: Understanding Business Cycles

1. Price index at time t_2 = $\frac{\text{Value of the Consumption Basket at } t_2}{\text{Value of the Consumption Basket at } t_1} \times 100$
Inflation Rate = $\left(\frac{\text{Price Index at time } t_2}{100} \right) - 1$

2. Fisher Index = $\sqrt{I_p \times I_L}$ (where, I_L = Laspeyres index and I_p = Paasche Index)

3. *Unit labor cost (ULC) indicator* = $\frac{\text{Total labor compensation per hour per worker}}{\text{Output per hour per worker}}$

4. Velocity of money = $\frac{\text{Nominal GDP}}{\text{Money Supply}}$

Reading 19: Monetary & Fiscal Policy

1. Total Money created = New deposit/ Reserve Req

2. Money Multiplier = $\frac{1}{\text{Reserve Req or reserve ratio}}$

3. Narrow money = M1 = currency held outside banks + checking accounts + traveller's check

4. Broad money = M2 = M1 + time deposits + saving deposits

5. M3 = M2 + deposits with *non-bank* financial institution

6. Quantity Theory of Money = $M \times V = P \times Y$ where,
M = Quantity of money
V = Velocity of circulation of money
P = Average price level
Y = Real output

7. Neutral Rate = Trend Growth + Inflation Target

8. Impact of Taxes and Government Spending: The Fiscal Multiplier
The net impact of the government sector on AD:

- $G - T + B$ = Budget surplus or Budget deficit
where, G = government spending, T = taxes, B = transfer benefits
- Disposable income = Income – Net taxes = $(1 - t)$ Income
where, Net taxes = taxes – transfer payments, t = net tax rate

9. Fiscal Multiplier (in the absence of taxes) = $1/(1 - \text{MPC})$

- $\text{MPS} = 1 - \text{MPC}$.
- Total increase in income and spending = Fiscal multiplier $\times$ G

10. Fiscal Multiplier (in the presence of taxes)

- $\text{MPC (with taxes)} = \text{MPC} \times (1 - t)$
- Fiscal multiplier = $\frac{1}{1 - \text{MPC} (1 - t)}$
- Total $\uparrow$ in income and spending = Fiscal multiplier $\times$ G
- Initial $\uparrow$ in consumption due to reduction in taxes = $\text{MPC} \times \text{tax cut amount}$
- Total or cumulative effect of tax cut = multiplier $\times$ initial change in consumption

11. Cumulative multiplier = $\frac{\text{cumulative effect on real GDP over the two years}}{\% \text{ of GDP}}$

Reading 20: International Trade & Capital Flows

1. Terms of trade = $\frac{\text{Price of exports}}{\text{Price of imports}}$

2. Terms of Trade (as an index number) = $\frac{\text{Avg price of exports}}{\text{Avg price of imports}}$

3. Net exports = Value of a country's (exports – imports)

4. Net welfare effect = consumer's surplus loss + producer's surplus gain + Govt. revenue
5. Closed Economy's output = $Y = C + I + G$
6. Open Economy's output = $Y = C + I + G + (X - M)$
 - Current Account Balance = $X - M = Y - C - I - G$
7. Consumption = Income + transfers – taxes – saving

$C = Y^d - S_p = Y + R - T - S_p$ And,
 $CA = S_p - I + \text{Govt surplus (or Govt saving)}$
 $= S_p - I + (T - G - R)S_p + S_g = I + CA$

where, S_g = Govt savings
 $S_p = I + CA - S_g$

 - Current Account Imbalance $CA = S_p + S_g - I$

Reading 21: Currency Exchange Rates

1. Foreign price level in domestic currency = $S_{d/f} \times P_f$
2. Real exchange rate_(D/f) = $(S_{d/f} \times P_f) / P_d = S_{d/f} \times (P_f / P_d)$
3. Real Exchange Rate_{domestic/foreign} = $S_{d/f} \times \left(\frac{CPI_f}{CPI_d} \right)$

4. Change in Real Exchange rate = $\left(1 + \frac{\Delta S_{d/f}}{S_{d/f}} \right) \times \frac{1 + \frac{\Delta P_f}{P_f}}{\left(1 + \frac{\Delta P_d}{P_d} \right)} - 1$
5. Direct Quote = $\frac{1}{\text{Indirect Quote}}$
6. Points on a forward rate quote = Fwd X-rate quote – Spot X-rate quote
7. Forward rate = Spot X-rate + $\frac{\text{Forward points}}{10,000}$
8. Forward premium/discount (in %) = $\frac{\text{spot X-rate} + (\text{forward points}/10,000)}{\text{spot X-rate}} - 1$
9. To convert spot rate into a forward quote (when points are represented as %) = Spot exchange rate $\times (1 + \% \text{ premium or discount})$
10. Arbitrage relationship is stated as follows:
 - $(1 + i_d) = S_{f/d} (1 + i_f) \left(\frac{1}{F_{f/d}} \right)$
 - In case of indirect quote, Arbitrage relationship is: $(1 + i_d) = (1/S_{f/d})(1 + i_f)F_{f/d}$
 - $F_{f/d} = S_{f/d} \left(\frac{1 + i_f}{1 + i_d} \right)$
 - Forward rate as a % of spot rate = $\frac{F_{f/d}}{S_{f/d}} = \left(\frac{1 + i_f}{1 + i_d} \right)$

11. Return on hedged foreign investment (with a quoted forward rate) = $S_{f/d} \left(1 + i_f \right) \left(\frac{1}{F_{f/d}} \right)$
12. Expected % change in the spot rate = $\frac{\hat{S}_{t+1}}{S_t} - 1 = \% \Delta \hat{S}_{t+1} = \left(\frac{i_f - i_d}{1 + i_d} \right)$
 - Forward points: $F_{f/d} - S_{f/d} = S_{f/d} \left(\frac{i_f - i_d}{1 + i_d} \right) \tau$ (where τ is quoted interest rate period)
13. Relationship between the trade balance and expenditure/ saving decisions:
 $= Ex - Im = (Sav - Inv) + (T - G)$

where T= taxes net of transfers
 G= government expenditures)
14. Price elasticity of demand = $\epsilon = \frac{\% \text{ change in quantity}}{\% \text{ change in price}} = \frac{\% \Delta Q}{\% \Delta P}$
15. Expenditure (R) = Price $\times$ Quantity = $P \times Q$
 - $\% \Delta \text{ in expenditure} = \% \Delta R = \% \Delta P + \% \Delta Q = (1 - \epsilon) \% \Delta P$
16. Basic idea of Marshall-Lerner condition = $\omega_x \epsilon_x + \omega_M (\epsilon_M - 1) > 0$ where,

ω_x = share of exports
 ϵ_x = price elasticity of foreign demand for domestic country exports

ω_M = share of imports

ϵ_M = price elasticity of domestic country demand for imports

17. Trade balance = Income (GDP) – Domestic expenditure = Absorption

Reading 22: Financial Statement Analysis: An Introduction

- Gross Profit = Revenue – Cost of sales
- Operating Profit or EBIT = Gross profit – Operating costs + Other operating income
- Profit before tax = EBIT – Interest expense
- Profit after tax = Profit before tax – Income tax expense

Reading 23: Financial Reporting Mechanics

- Owner's Equity = Contributed Capital + R.E
- End R.E = Beg R.E + Net income – Dividends
- Assets = Liabilities + Contributed Capital + Beg R.E + Revenue – Expenses – Dividends

Reading 24: Financial Reporting Standards

Reading 25: Understanding Income Statements

- Revenue recognized on Prorated basis =
$$\frac{\text{Total Amount of Cost}}{\text{Time of the contract}}$$
- Revenue recognized under Percentage-of-Completion Method = % of Total cost spent by the firm $\times$ Total Contract Revenue
- Revenue recognized when outcome cannot be reliably measured = Contract costs incurred
- Revenue recognized under installment method = $\left(\frac{\text{Profit}}{\text{Sales}}\right) \times \text{Cash receipt}$
- Wgtd Avg cost per unit =
$$\frac{\text{Total Cost of Goods available for Sale}}{\text{Total units available for Sale}}$$
- COGS using Wgtd Avg Cost = No of units sold $\times$ Wgtd Avg cost per unit
- COGS using LIFO = Total cost – Value of ending inventory
- Annual Depreciation Expense (using Straight-Line Method) =
$$\frac{\text{Cost} - \text{Residual Value}}{\text{Estimated Useful Life}}$$
- Annual Depreciation Expense (Declining balance method) =
$$\frac{100\%}{\text{Useful life}} \times \text{Acceleration factor (say 200\% or 2)} \times \text{Net Book Value}$$

$$10. \text{ Basic EPS} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Wght Avg No of shares outstanding}}$$

$$11. \text{ Diluted EPS for preferred stock} = \frac{\text{Net Income}}{\left(\text{Wght Avg No of shares o/s} + \text{New common shares that would have been issued at conversion} \right)}$$

$$12. \text{ Diluted EPS for convertible debt} = \frac{\text{Net income} + \text{AT i on convertible debt} - \text{Preferred Div}}{\left(\text{Wght Avg of shares o/s} + \text{Additional common shares that would have been issued at conversion} \right)}$$

$$13. \text{ Diluted EPS using Treasury Stock Method} = \frac{(\text{Net Income} - \text{Preferred dividends})}{[\text{Wght Avg of shares} + (\text{New shares at option exercise} - \text{Shares purchased with Cash received upon exercise}) \times (\text{Proportion of Yr})]}$$

$$14. \text{ Net Profit Margin} = \frac{\text{Net Income}}{\text{Revenue}}$$

$$15. \text{ Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Revenue}}$$

$$16. \text{ Comprehensive EPS} = \text{EPS} + \text{Other Comprehensive Income per share}$$

Reading 26: Understanding Balance Sheets

- Percentage of A/C Receivable estimated to be uncollectible =
$$\frac{\text{Allowance for Doubtful A/C}}{\text{Gross amount of A/C Receivable}}$$
- Net Identifiable Assets = Fair value of identifiable assets – Fair value of liabilities & contingent liabilities

3. Amortized cost of PPE = Historical cost – Accumulated depreciation – Impairment losses
4. Carrying value for PPE under revaluation model
= Fair value at date of revaluation – Accumulated depreciation (if any)
5. Amortized cost of PPE = Historical cost – Accumulated depreciation – Impairment losses
6. Carrying value for PPE under revaluation model
= Fair value at date of revaluation – Accumulated depreciation (if any)
7. Deferred tax liability = Taxable income < Reported Financial Statement Income before taxes
8. Deferred tax liability = Actual income tax payable in a period < Income tax expense
9. Vertical common-size balance-sheet =
$$\frac{\text{Balance sheet Amount}}{\text{Total Assets}}$$
10. Current ratio =
$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$
11. Quick (acid test) =
$$\frac{\text{Cash+Marketable securities+Receivables}}{\text{Current Liabilities}}$$

$$12. \text{Cash ratio} = \frac{\text{Cash+Marketable securities}}{\text{Current Liabilities}}$$

$$13. \text{Long-term debt-to-equity} = \frac{\frac{\text{Total long-term debt}}{\text{Total Equity}}}{\text{Total Equity}}$$

$$14. \text{Debt-to-Equity} = \frac{\text{Total Debt}}{\text{Total Equity}}$$

$$15. \text{Total Debt} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

$$16. \text{Financial Leverage} = \frac{\text{Total Assets}}{\text{Total Equity}}$$

Reading 27: Understanding Cash Flow Statements

1. End Cash = Beg cash + Cash receipts (from operating, investing, and financing activities) – Cash payments (for operating, investing, and financing activities)
2. End A/c Receivable = Beg A/c Receivable + Revenues – Cash collected from customers
3. Cash received from customers = Revenue – Increase in a/c receivable
4. Purchases from suppliers = COGS + Increase in inventory
5. Cash paid to suppliers = Cogs + Increase in inventory – Increase in a/c payable

6. End Inventory = Beg inventory + Purchases – COGS
7. End a/c payable = Beg a/c payable + Purchases – Cash paid to suppliers
8. Cash paid to employees = Salary and wages expense – Increase in salary and wages payable
9. End salary and wages payable = Beg salary and wages payable + Salary and wages expense – cash paid to employees
10. Cash paid for other operating expenses = Other operating expenses – Decrease in prepaid expenses – Increase in other accrued liabilities
11. Cash paid for interest = Interest expense + Decrease in interest payable
12. End Interest Payable = Beg interest payable + Interest expense – Cash paid for interest
13. Cash paid for income taxes = Income tax expense – Increase in income tax payable
14. Historical cost of equipment sold = Beg balance equipment + Equipment purchased – End balance equipment

$$15. \text{ Accumulated Dep on equipment sold} = \text{Beg balance accumulated dep} + \text{Dep expense} - \text{End balance accumulated dep}$$

$$16. \text{ Cash received from sale of equipment} = \text{Historical cost of equipment sold} - \text{Accumulated dep on equipment sold} + \text{gain on sale of equipment}$$

$$17. \text{ Dividends paid} = \text{Beg balance of R.E} + \text{Net income} - \text{End balance of R.E}$$

$$18. \text{ FCFF} = \text{Net income} + \text{Non-cash charges} + \text{Interest expense} (1 - \text{tax rate}) - \text{Cap exp} - \text{WC expenditures}$$

$$19. \text{ FCFF} = \text{CFO} + \text{Interest expense} (1 - \text{Tax rate}) - \text{Cap exp}$$

$$20. \text{ FCFE} = \text{CFO} - \text{Cap exp} + \text{Net borrowing}$$

$$21. \text{ CF to revenue} = \frac{\text{CFO}}{\text{Net Revenue}}$$

$$22. \text{ Cash ROA} = \frac{\text{CFO}}{\text{Average Total Assets}}$$

$$23. \text{ Cash ROE} = \frac{\text{CFO}}{\text{Average shareholders' equity}}$$

$$24. \text{ Cash to income} = \frac{\text{CFO}}{\text{Operating income}}$$

$$25. \text{ Cash flow per share} = \frac{(\text{CFO} - \text{Preferred Dividends})}{\text{No of common shares o/s}}$$

$$26. \text{ Debt Coverage} = \frac{\text{CFO}}{\text{Total Debt}}$$

$$27. \text{ Interest Coverage} = \frac{(\text{CFO} + \text{Interest paid} + \text{Taxes paid})}{\text{Interest paid}}$$

$$28. \text{ Reinvestment} = \frac{\text{CFO}}{\text{Cash paid for long-term assets}}$$

$$29. \text{ Debt payment} = \frac{\text{CFO}}{\text{Cash paid for LT debt repayment}}$$

$$30. \text{ Dividend payment} = \frac{\text{CFO}}{\text{Dividends paid}}$$

$$31. \text{ Investing and Financing} = \frac{\text{CFO}}{\text{Cash outflows for investing and financing activities}}$$

Reading 28: Financial Analysis Techniques

$$1. \text{ Compound Growth Rate} = \left[\left(\frac{\text{End Value}}{\text{Beg Value}} \right)^{\left(\frac{1}{\text{No of periods}} \right)} \right] - 1$$

$$2. \text{ Combined ratio} = \frac{\text{Losses and Expenses}}{\text{Net Premiums Earned}}$$

$$3. \text{ Operating ROA} = \frac{\text{Operating Income}}{\text{Avg Total Assets}}$$

$$4. \text{ ROA} = \frac{\text{Net Income}}{\text{Avg Total Assets}} \text{ or } \text{ROA} = \frac{\text{Net Income} + \text{Interest Expense} (1 - \text{Tax rate})}{\text{Avg Total Assets}}$$

$$5. \text{ Effective Tax Rate} = \frac{\text{Income Tax}}{\text{Earnings before Tax}}$$

$$6. \text{ Vertical common size income statement} = \frac{\text{Income statement Item}}{\text{Revenue}}$$

$$7. \text{ Horizontal common size balance sheet} = \frac{\text{Balance sheet item in Year 2}}{\text{Balance sheet item in Year 1}}$$

$$8. \text{ Inventory turnover} = \frac{\text{Cost of sales or cost of goods sold}}{\text{Avg Inventory}}$$

$$9. \text{ Days of Inventory on Hand (DOH)} = \frac{\text{No of Days in period}}{\text{Inventory Turnover}}$$

$$10. \text{ Receivables Turnover} = \frac{\text{Revenue}}{\text{Avg Receivables}}$$

$$11. \text{ Days of Sales Outstanding (DSO)} = \frac{\text{No of Days in Period}}{\text{Receivables turnover}}$$

$$12. \text{ Avg A/c Receivable Balance} = \text{Avg Days' Credit Sales} \times \text{DSO or } \text{Avg A/c Receivable Balance} = \frac{\text{Sales}}{\text{Turnover}} = \frac{\text{Sales}}{\frac{365}{\text{DSO}}}$$

$$13. \text{ Payables turnover} = \frac{\text{Purchases}}{\text{Avg trade payables}}$$

$$14. \text{ No of Days of Payables} = \frac{\text{No of Days in period}}{\text{Payables Turnover}}$$

$$15. \text{ WC Turnover} = \frac{\text{Revenue}}{\text{Avg WC}}$$

$$16. \text{ Fixed Asset Turnover} = \frac{\text{Revenue}}{\text{Avg Net Fixed Assets}}$$

$$17. \text{ Total Asset Turnover} = \frac{\text{Revenue}}{\text{Avg Total Assets}}$$

$$18. \text{ Pretax margin} = \frac{\text{Earnings before tax but after interest}}{\text{Revenue}}$$

$$19. \text{ Return on Total Capital} = \frac{\text{EBIT}}{\text{Short and long term debt and equity}}$$

$$20. \text{ ROE} = \frac{\text{Net Income}}{\text{Avg Total Equity}}$$

- $\text{ROE} = \text{ROA} \times \text{Leverage}$
- $\text{ROE} = \text{Tax Burden} \times \text{Interest Burden} \times \text{EBIT Margin} \times \text{Total Asset Turnover} \times \text{Leverage}$

$$21. \text{ Return on Common Equity} = \frac{\text{Net Income} - \text{Preferred Dividends}}{\text{Avg Common Equity}}$$

$$22. \text{ Coefficient of Variation of Operating Income} = \frac{\text{S.D of Operating Income}}{\text{Avg Operating Income}}$$

$$23. \text{ Coefficient of Variation of Net Income} = \frac{\text{S.D of Net Income}}{\text{Avg Net Income}}$$

$$24. \text{ Coefficient of Variation of Revenues} = \frac{\text{S.D of Revenue}}{\text{Avg Revenue}}$$

$$25. \text{ Monetary Reserve Requirement (Cash Reserve Ratio)} = \frac{\text{Reserves held as Central Bank}}{\text{Specified Deposit Liabilities}}$$

$$26. \text{ Liquid Asset Requirement} = \frac{\text{Readily Marketable Securities}}{\text{Specified Deposit Liabilities}}$$

$$27. \text{ Net Interest Margin} = \frac{\text{Net Interest Income}}{\text{Total Interest Earning Assets}}$$

$$28. \text{ Sales per Square Meter} = \frac{\text{Revenue}}{\text{Total Retail Space in Square Meters}}$$

$$29. \text{ Average Daily Rate} = \frac{\text{Room Revenue}}{\text{No of Rooms sold}}$$

$$30. \text{ Occupancy Rate} = \frac{\text{No of Rooms Sold}}{\text{No of Rooms available}}$$

$$31. \text{ EBIT Interest Coverage} = \frac{\text{EBIT}}{\text{Gross Interest}}$$

$$32. \text{ EBITDA Interest Coverage} = \frac{\text{EBITDA}}{\text{Gross Interest}}$$

$$33. \text{ FFO Interest Coverage} = \frac{\text{FFO} + \text{Interest Paid} - \text{Operating Lease Adjustments}}{\text{Gross Interest}}$$

$$34. \text{ Return on Capital} = \frac{\text{EBIT}}{\text{Avg Capital}} = \frac{\text{EBIT}}{\text{Avg (Equity} + \text{Non current deferred taxes} + \text{debt)}}$$

$$35. \text{ FFO to Debt} = \frac{\text{FFO}}{\text{Total Debt}}$$

$$36. \text{ Free Operating CF to Debt} = \frac{\text{CFO} - \text{Cap Exp}}{\text{Total Debt}}$$

$$37. \text{ Discretionary CF to Debt} = \frac{\text{CFO} - \text{Cap exp} - \text{Dividends paid}}{\text{Total debt}}$$

$$38. \text{ Net CF to Capital expenditures} = \frac{\text{FFO} - \text{Dividends}}{\text{Cap exp}}$$

$$39. \text{ Debt to EBITDA} = \frac{\text{Total debt}}{\text{EBITDA}}$$

$$40. \text{ Total Debt to total debt plus Equity} = \frac{\text{Total debt}}{\text{Total debt} + \text{Equity}}$$

$$41. \text{ Z-Score} = 1.2 \times \left(\frac{\text{CA} - \text{CL}}{\text{TA}} \right) + 1.4 \times \left(\frac{\text{RE}}{\text{TA}} \right) + 3.3 \times \left(\frac{\text{EBIT}}{\text{TA}} \right) + 0.6 \times \left(\frac{\text{MV of stock}}{\text{BV of liabilities}} \right) + 1.0 \times \left(\frac{\text{Sales}}{\text{TA}} \right)$$

$$42. \text{ Segment margin} = \frac{\text{Segment Profit (Loss)}}{\text{Segment Revenue}}$$

$$43. \text{ Segment turnover} = \frac{\text{Segment Revenue}}{\text{Segment Assets}}$$

$$44. \text{ Segment ROA} = \frac{\text{Segment Profit (Loss)}}{\text{Segment Assets}}$$

$$45. \text{ Segment Debt Ratio} = \frac{\text{Segment Liabilities}}{\text{Segment Assets}}$$

Reading 29: Inventories

1. $\text{NRA} = \text{Estimated Selling Price} - \text{Estimated Costs of completion and disposal}$

2. Inventory amount net of valuation allowance = Carrying amount of Inventory – Write downs
3. $(\text{NRA} - \text{Normal Profit Margin}) \leq \text{MV} \leq \text{NRA}$

Reading 30: Long-Lived Assets

1. Dep Exp under Straight-line Method =
$$\frac{\text{Depreciable Cost}}{\text{Estimated Useful Life}} = \frac{\text{Historical Cost} - \text{Estimated Residual (salvage) Value}}{\text{Estimated Useful Life}}$$
2. Dep Exp under Units-of-Production Method =
$$\text{Depreciable Cost} \times \frac{\text{Production in the Period}}{\text{Estimated Productive Capacity}}$$
3. Carrying amount under cost model = Historical Cost – Accumulated Dep or Amortization
4. Carrying amount under revaluation model = Fair value at the date of revaluation – Any subsequent Accumulated Dep or Amortization
5. Impairment Loss (IFRS) = Recoverable Amount – Net Carrying Amount

Where, Recoverable amount = Max [(Fair value – Costs to sell); Value in Use] and Value in use = PV of Expected Future CFs

6. Impairment Loss (US GAAP) = Asset's Fair Value – Carrying Amount If Carrying amount > Undiscounted Expected Future Cash Flows

Reading 31: Income Taxes

1. Deferred tax asset = Company's taxable income > Accounting profit
2. Tax base of revenue received in advance = Carrying amount – Any amount of revenue that will not be taxed at a future date
3. Reported Effective Tax Rate =
$$\frac{\text{Income Tax expense}}{\text{Pre tax income or Accounting Profit}}$$
4. Deferred tax liability = Carrying amount of asset > Tax base of asset
5. Deferred tax asset = Carrying amount of asset < Tax base of asset
6. Deferred tax asset = Carrying amount of liability > Tax base of asset
7. Deferred tax liability = Carrying amount of liability < Tax base of asset
8. Company's tax expense (or credit) reported on its income statement = Income tax liability currently payable + Δ in deferred tax asset / liability
Where,

- Income Tax liability currently payable = Taxable income $\times$ Tax rate
- Δ in deferred tax asset / liability = Diff b/w the balance of the deferred tax asset / liability for the current period and the balance of the previous period.

9. The company's tax expense (or credit) reported on its income statement = Taxes payable + (Δ Deferred tax liability - Δ Deferred tax asset)
Where,
 - Income Tax liability currently payable = Taxable income $\times$ Tax rate
 - Deferred tax liability = (carrying amount – tax base) $\times$ tax rate
 - Deferred tax asset = (tax base – carrying amount) $\times$ tax rate
10. Tax base of a liability = Carrying amount of the liability – Amounts that will be deductible for tax purposes in the future

Reading 32: Non-current (Long-term) Liabilities

1. Annual Interest Payment = Face Value $\times$ Coupon Rate
2. Sale proceeds of bond = Sum of PV of Interest Payments + PV of Face value of Bond

3. When Face value - Sale proceed is > zero, discount
4. When Face value – Sale proceed is < zero, premium
5. Bond payable = Face value – (+) Discount (Premium)
6. Total Interest Expense (in case of discount) = Periodic interest payments + Amortization of Discount
7. Total Interest Expense (in case of premium) = Periodic interest payments - Amortization of Premium
8. Amount of Bonds payable reported on the balance sheet = Historical cost +/- Cumulative amortization (or amortization cost)
9. Amount of Bonds payable initially reported on the balance sheet under IFRS = Sales proceeds – Issuance costs
10. Amount of Bonds payable initially reported on the balance sheet under US GAAP = Sales proceeds
11. Bond *i*-exp. under effective *i*-rate method = Carrying value of the bonds at the beg. of the period × Effective *i*-rate
12. Bond Interest Payment under effective interest rate method = Face value of the bonds × Contractual (coupon) rate
13. Amortization of the discount or premium under effective interest rate method = Bond interest expense – Bond interest payment
14. Bond Discount/Premium Amortization under Straight-line Method =
$$\frac{\text{Bond Discount or premium}}{\text{No of Interest Periods}}$$
15. No of shares subscribed when warrants are exercised =
$$\frac{\text{Aggregate principal amount of debt}}{\text{Par value of a lot}} \times \text{shares subscribed per lot}$$
16. Carrying amount of the leased asset = Initial recognition amount – Accumulated depreciation
17. Accumulated depreciation = Prior year's accumulated depreciation + Current year's depreciation expense
18. Interest expense = Lease liability at the beg of the period × interest rate implicit in the lease
19. Sales revenue = lower of the fair value of the asset and PV of the min lease payments
20. Cost of sales = Carrying amount of the leased asset – PV of the estimated unguaranteed residual value
21. Interest Revenue = Lease receivable at the beg of the period × Interest rate
22. Net interest expense = Beg Net pension liability × Discount rate
23. Net Interest income = Beg Net Pension asset × Discount rate
24. Reported pension expense = Pension costs – Expected return on Pension plan assets
25. Funded Status = PV of the Defined benefit obligations – Fair value of the plan assets

Reading 33: Financial Reporting Quality

Reading 34: Financial Statement Analysis: Applications

1. Company's sales = Projected market share × Projected total industry sales
2. Forecast amount of profit for a given period = Forecasted amount of sales × Forecast of the selected profit margin
3. Retained CF (RCF) / Total debt =
$$\frac{(\text{operating CF before WC changes} - \text{dividends})}{\text{total debt}}$$
4.
$$\frac{\text{Retained CF} - \text{Cap exp}}{\text{Total Debt}}$$

5. Inventory value adjusted to FIFO basis =
End Inventory value under LIFO + End LIFO reserve balance
6. COGS adjusted to a FIFO basis = COGS under LIFO – (End LIFO reserve – Beg LIFO reserve)
7. Useful life of the company's overall asset base that has passed = $\frac{\text{Accumulated Dep}}{\text{Gross PPE}}$
8. Avg age of the asset base = $\frac{\text{Accumulated Dep}}{\text{Annual Dep expense}}$
9. Remaining useful life of the asset = $\frac{\text{Net PPE (net of accumulated dep)}}{\text{Annual dep expense}}$
10. Avg depreciable life of the assets at installation = $\frac{\text{Gross PPE}}{\text{Annual Dep expense}}$
11. % of asset base that is being renewed through new capital investment = $\frac{\text{Capex}}{\text{Gross PPE} + \text{Capex}}$
12. Adjusted BV = Total stockholders' equity – Goodwill
13. Adjusted Price to BV ratio = $\frac{\text{Price (market capitalization)}}{\text{Adjusted BV}}$
14. Tangible B.V = Total stockholders' equity – Goodwill – Other intangible assets

15. Price to tangible BV ratio = $\frac{\text{Price}}{\text{Tangible BV}}$
16. Adjusted debt-to-equity ratio = $\frac{\text{Reported debt} + \text{PV of operating lease}}{\text{Reported Equity}}$
17. Adjusted debt-to-asset ratio = $\frac{\text{Reported debt} + \text{PV of operating lease}}{\text{Reported Asset} + \text{PV of operating lease}}$
18. Adjusted Asset Turnover ratio = $\frac{\text{Sales}}{\text{Reported Avg total assets} + \text{PV of operating lease}}$
19. PV of future operating lease payments = $\frac{\text{PV of capital lease payments}}{\text{Total Capital Lease payments}} \times \text{Total Future Operating Lease Payments}$
20. Interest expense = Interest $\times$ PV of the lease payments
21. Depreciation expense estimated on straight-line basis = $\frac{\text{PV of the lease payments}}{\text{No of yrs of future lease payments}}$
22. Adjusted Interest Coverage ratio = $\frac{(\text{EBIT}) + \text{rent exp} * - \text{Dep exp} *}{i \text{ payments} + i \text{ expense} *}$

* associated with the operating lease obligations

Reading 35: Capital Budgeting

1. Incremental CF = CF with a decision - CF without that decision

$$2. \text{ NPV} = \text{PV of cash inflows} - \text{IO} = \sum_{t=1}^n \frac{\text{AT CFs at time } t}{(1 + \text{Req RoR})^t} - \text{IO}$$

$$3. \text{ Avg Accounting RoR (AAR)} = \frac{\text{Avg NI after dep \& taxes (before interest)}}{\text{Avg BV of Invst}}$$

$$4. \text{ PI} = \frac{\text{PV of future CFs}}{\text{IO}} = 1 + \frac{\text{NPV}}{\text{IO}}$$

5. Value of a company = Value of company's existing invst + Net PV of all of company's future invst

Reading 36: Cost of Capital

$$1. \text{ WACC} = w_d r_d (1 - t) + w_p r_p + w_e r_e$$

$$2. \text{ Debt-to-Equity Ratio conversion into weight (i.e. Debt / (Debt + Equity))} = \frac{\frac{\text{Debt}}{\text{Equity}}}{\left(1 + \frac{\text{Debt}}{\text{Equity}}\right)}$$

3. Optimal Capital Budget is the point where MC of capital = Marginal return from investing

$$4. \text{ After-tax cost of debt} = \text{Before-tax Marginal Cost of Debt} \times (1 - \text{firm's marginal tax rate})$$

$$5. \text{ Preferred Stock Price per Share} = \frac{\text{Pref Stock Div per Share}}{\text{Cost of Pref Stock}}$$

6. Expected Return on Stock I (under CAPM)
 $= E(R_i) = R_F + \beta_i [E(R_M) - R_F]$

7. Expected Return on Stock I $= E(R_i) = R_F + \beta_{i1} (\text{Factor risk premium})_1 + \beta_{i2} (\text{Factor risk premium})_2 + \dots + \beta_{ij} (\text{Factor risk premium})_j$

8. Cost of Equity $= r_e = \frac{D_1}{P_0} + g$

9. Expected Growth Rate of Dividends

$$g = \left(1 - \frac{D}{EPS}\right) \times ROE$$

$$g = \text{retention rate} \times ROE$$

10. Company's stock returns $= R_{it} = \hat{a} + \hat{b}R_{mt}$

11. Unlevered β of Comparable Company =

$$\beta_{U, compa} = \frac{\beta_{L, comparable}}{1 + \left\{ (1 - t_{comparable}) \frac{D_{comparable}}{E_{comparable}} \right\}}$$

12. Levered β of Project =

$$\beta_{L, proj} = \beta_{U, comp} \left[1 + \left\{ (1 - t_{proj}) \frac{D_{proj}}{E_{proj}} \right\} \right]$$

13. $\beta_{asset} = \frac{\beta_{equity}}{1 + \left\{ (1 - t) \frac{D}{E} \right\}}$

14. $\beta_{equity} = \beta_{asset} \left[1 + \left\{ (1 - t) \frac{D}{E} \right\} \right]$

15. Sovereign yield spread = Govt bond yield (denominated in developed country's currency) – T.B yield on a similar maturity bond in developed country

16. Country equity premium = Sovereign yield spread $\times \left[\frac{\text{Ann S.D of Equity index}}{\text{Ann S.D of sovereign bond Mkt in terms of developed mkt currency}} \right]$

17. Cost of equity $= K_e = R_F + \beta [(E(R_M) - R_F) + CRP]$

18. Breakpoint =

$$\frac{\text{Amount of capital at which source's cost of cap } \Delta}{\text{Prop of new cap raised from the source}}$$

19. Cost of Capital (hen flotation costs are in monetary terms $= r_e = \left(\frac{D_1}{P_0 - F} \right) + g$

20. When FC are in terms of % of the share price: Cost of Equity $= r_e = \left(\frac{D_1}{P_0 - F} \right) + g$

21. If FC are not tax deductible: NPV = PV of Cash Inflows – IO – (FC in % $\times$ New Equity Capital)

22. If FC are tax deductible: NPV = PV of Cash Inflows – IO – [(FC in % $\times$ New Equity Capital) $\times$ (1 – Marginal Tax Rate)]

23. Asset β = (Debt β $\times$ Proportion of Debt) + (Equity β $\times$ Proportion of Equity)

Reading 37: Measures of Leverage

1. Contribution Margin (CM) = (# of units sold) $\times$ [(price per unit) – (variable cost per unit)]

2. Per unit CM = Price per unit – Variable cost per unit

3. Operating income = CM – Fixed Operating Costs

4. $DOL = \frac{\% \Delta \text{ in Operating Income (EBIT)}}{\% \Delta \text{ in Units Sold}}$

or
 $DOL = \frac{CM}{CM - \text{Fixed Operating Cost}}$

5. $DFL = \frac{\% \Delta \text{ in Net Income}}{\% \Delta \text{ in Operating Income}}$ or
 $\frac{CM - \text{Fixed Op Cost}}{(CM - \text{Fixed Op Cost}) - \text{Fixed Fin Cost}}$

6. $DTL = \frac{\% \Delta \text{ in Net Income}}{\% \Delta \text{ in No of Units Sold}} = DOL \times DFL = \frac{CM}{CM - \text{Fixed Op Cost} - \text{Fixed Fin Cost}}$

7. Break-even Revenue = (Variable cost per unit $\times$ Break-even Number of Units) + Fixed Operating costs + Fixed Financial Cost

8. Breakeven Number of units $= Q_{BE} = \frac{\text{Fixed Operating Costs} + \text{Fixed Financial Costs}}{\text{Price per unit} - \text{Variable cost per unit}}$

9. Operating Breakeven $= Q_{OBE} = \frac{\text{Fixed Operating Cost}}{\text{Price per unit} - \text{Variable cost per unit}}$

Reading 38: Dividends & Share Repurchases: Basics

- Company's payout for the year = Cash dividends + Value of shares repurchased in any given year
- Dividend Payout ratio =
$$\frac{\text{Common share cash dividends}}{\text{Net Income available to common shares}}$$
- EPS after Dividend = EPS before Dividend $\times \frac{\text{Shares o/s before Dividend}}{\text{Shares o/s after Dividend}}$
- Stock Price after Dividend = Stock Price before Dividend $\times$ EPS after Dividend
- Total Market Value after Dividend = Shares outstanding after Dividend $\times$ Stock price after Dividend
- Stock price after 2-for-1 stock split =
$$\frac{\text{Stock price before stock split}}{2}$$
- EPS after 2-for-1 stock split =
$$\frac{\text{EPS before stock split}}{2}$$
- DPS after 2-for-1 stock split =
$$\frac{\text{DPS before stock split}}{2}$$
- EPS after buyback =
$$\frac{(\text{Earnings} - \text{After tax Cost of Funds})}{\text{Shares Outstanding after Buyback}}$$

- Ex-dividend value of share = Stock price – Dividend per share
- Market value of Equity after distribution of cash dividends =
$$\frac{[(\# \text{ of shares o/s}) \times (\text{MV share}) - \text{Cash div}]}{\# \text{ of shares o/s}}$$
- Post-repurchase share price =
$$\frac{[(\# \text{ of shares o/s}) \times (\text{MV share}) - \text{worth of Share repurchase}]}{((\# \text{ of shares o/s}) - \# \text{ of shares (can be repurchased by a Co)})}$$

Reading 39: Working Capital Management

- Operating cycle = No of days of inventory + No of days of receivables
- Net operating cycle = No of days of inventory + No of days of receivables – No of days payables
- Money Market Yield =
$$\left(\frac{\text{Face value} - \text{Purchase price}}{\text{Purchase price}} \right) \times \left(\frac{360}{\text{No of days to maturity}} \right)$$
- Bond Equivalent Yield =
$$\left(\frac{\text{Face value} - \text{Purchase price}}{\text{Purchase price}} \right) \times \left(\frac{365}{\text{No of days to maturity}} \right)$$
- Discount-basis Yield =
$$\left(\frac{\text{Face value} - \text{Purchase price}}{\text{Face Value}} \right) \times \left(\frac{360}{\text{No of days to maturity}} \right)$$

- Wght Avg collection period = wghts $\times$ Avg no of days to collect accounts within each aging category

Where, Weights = % of total receivables in each category

- Float Factor =
$$\frac{\text{Avg Daily Float}}{\text{Avg Daily Deposit}} = \frac{\text{Avg Daily Float}}{\frac{\text{Total Amount of Checks Deposited}}{\text{No of Days}}}$$

Where, Float = Amount of money that is in transit b/w payments (by customers) and funds (usable by co)

- Value of stretching payment = A/c payable $\times$ Co's opportunity cost for ST funds
- Cost of Trade Credit =
$$\left(1 + \frac{\text{Discount}}{1 - \text{Discount}} \right)^{\left(\frac{365}{n} \right)} - 1$$
 where n = days beyond discount period
- Cost of Line of Credit =
$$\frac{\text{Interest} + \text{Commitment fee}}{\text{Loan Amount}}$$
- Bankers Acceptance Cost =
$$\frac{\text{Interest}}{\text{Net proceeds}} = \frac{\text{Interest}}{\text{Loan amount} - \text{Interest}}$$
- Commercial Paper Cost =
$$\frac{\text{Interest} + \text{Dealer's commission} + \text{Backup costs}}{\text{Loan amount} - \text{Interest}}$$
- Annualized cost = Cost $\times$ 12

Reading 40: The Corporate Governance of Listed Companies

Reading 41: Portfolio Management: An Overview

- NAV of bond mutual fund = $\frac{\sum(\text{value of each bond in the portfolio})}{\text{No of shares}}$
- New Shares that need to be created = $\frac{\text{Amount to be Invested in the Fund}}{\text{NAV or Total value of a Mutual Fund}}$
- New NAV of the Fund = NAV or Total value of a Mutual Fund + Amount to be invested in the Fund
- No of shares need to be retired = $\frac{\text{Amount to be withdrawn from the Fund}}{\text{NAV or Total value of a Mutual Fund}}$

Reading 42: Risk Management: An Introduction

Reading 43: Portfolio Risk & Return: Part I

- Total Return = Capital Gain (or Loss) + Dividend Yield
- Capital Gain = $\frac{P_t - P_{t-1}}{P_{t-1}}$
- Dividend Yield = $\frac{D_T}{P_0} - 1$

$$4. \text{ 3-Yr HPR} = [(1 + R_1) \times (1 + R_2) \times (1 + R_3)]^{1/3} - 1$$

$$5. \text{ Arithmetic mean (AM) } R = \bar{R}_i = \frac{R_{i1} + R_{i2} + \dots + R_{iT-1} + R_{iT}}{T} = \frac{1}{T} \sum_{t=1}^T R_{it}$$

$$6. \text{ Geometric R for n periods} = \bar{R}_{Gi} = [(1 + R_1)(1 + R_2) \dots (1 + R_n)]^{1/n} - 1$$

$$7. \text{ IRR} = \sum_{t=0}^T \frac{\text{CF at Time } t}{(1 + \text{IRR})^t} = 0$$

8. Annual Return (Ann R):

- Ann R = $(1 + \text{Quarterly R})^4 - 1$
- Ann R = $(1 + \text{Monthly R})^{12} - 1$
- Ann R = $(1 + \text{Weekly R})^{52} - 1$
- Ann R = $(1 + \text{Daily R})^{365} - 1$
- Weekly R = $(1 + \text{Daily R})^5 - 1$
- Weekly R = $(1 + \text{Annual R})^{1/52} - 1$

$$9. \text{ Portf R (for Two Assets)} = (\text{Wght of Asset 1} \times R \text{ of Asset 1}) + (\text{Wght of Asset 2} \times R \text{ of Asset 2})$$

$$10. \text{ Gross R} = R - \text{Trading exp} - \text{other exp directly related to the generation of returns.}$$

$$11. \text{ Net R} = \text{Gross R} - \text{All managerial and administrative exp}$$

$$12. \text{ After-tax nominal R} = \text{Total R} - \text{Any allowance for taxes on realized gains}$$

$$13. (1 + \text{Nominal R}) = (1 + \text{Real Rf R}) \times (1 + \text{Inf}) \times (1 + \text{RP})$$

$$14. (1 + \text{Real R}) = (1 + \text{Real Rf R}) \times (1 + \text{RP})$$

$$15. (1 + \text{Real R}) = \frac{(1 + \text{Nominal R})}{(1 + \text{Inf})}$$

$$16. \text{ Var of a Single Asset} = \sigma^2 = \frac{\sum_{t=1}^T (R_t - \mu)^2}{T}$$

$$17. \text{ Sample Variance} = s^2 = \frac{\sum_{t=1}^T (R_t - \bar{R})^2}{T-1}$$

$$18. \text{ Cov of R b/w two assets} = \text{Cov}(R_i, R_j) = \rho_{ij} \times \sigma_i \times \sigma_j$$

$$19. \text{ Portfolio Var} = \sigma_p^2 = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + 2\omega_1 \omega_2 \text{Cov}(R_1, R_2) = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + 2\omega_1 \omega_2 \rho_{12} \sigma_1 \sigma_2$$

$$20. \text{ Portfolio S.D.} = \sqrt{\text{Portfolio Variance}}$$

$$21. \text{ Cov b/w asset 1 \& asset 2} = \text{Correlation of Return b/w two assets} \times \text{S.D. of asset 1} \times \text{S.D. of asset 2}$$

$$22. \text{ Correlation of Return b/w two assets} = \frac{\text{Covariance of Return b/w two assets}}{\text{S.D. of asset 1} \times \text{S.D. of asset 2}}$$

$$23. 1 + \text{Expected Return} = 1 + E(R) = (1 + r_{rf}) \times [1 + E(\pi)] \times [1 + E(RP)]$$

$$24. \text{ Utility of an Invest} = \text{Expected Return} - \left(\frac{1}{2} \times \text{Risk Aversion Coefficient} \times \text{Var of Invest} \right)$$

$$25. \text{ Expected R of Portfolio} = E(R_p) = \omega_1 R_f + (1 - \omega_1)E(R_i)$$

$$26. \text{ Risk of Portfolio} = \sigma_p^2 = \omega_1^2 \sigma_f^2 + (1 - \omega_1)^2 \sigma_i^2 + 2\omega_1(1 - \omega_1)\rho_{12}\sigma_f\sigma_i = (1 - \omega_1)^2 \sigma_i^2 \text{ \& } \sigma_p = (1 - \omega_1) \sigma_i$$

$$27. \text{ Capital Allocation Line (CAL)} = E(R_p) = R_f + \frac{E(R_i) - R_f}{\sigma_i} \sigma_p$$

$$28. \text{ Portfolio Risk} = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + 2\omega_1\omega_2 \text{Cov}(R_1, R_2)$$

$$29. \text{ In portfolio of many asset} =$$

- $E(R_p) = \sum_{i=1}^N \omega_i E(R_i)$
- $\sigma_p^2 = \frac{\bar{\sigma}^2}{N} + \frac{(N-1)}{N} \overline{\text{Cov}}$

- $\sigma_p = \sqrt{\frac{\sigma^2}{N} + \frac{(N-1)}{N} \rho \sigma^2}$

$$30. \text{ New Asset should be included in the Portf only if } \frac{E(R_{\text{new}}) - R_f}{\sigma_{\text{new}}} > \frac{E(R_p) - R_f}{\sigma_p} \times \rho_{\text{new,p}}$$

Reading 44: Portfolio Risk & Return: Part II

1. Total Risk = Systematic risk + Nonsystematic risk = $\beta_i^2 \sigma_m^2 + \sigma^2 e$
2. Total risk of for a well-diversified portfolio = Systematic risk = $\beta_i \times \sigma_m$
3. Multi-Factor Model: $E(R_i) - R_f = \sum_{j=1}^k \beta_{ij} E(F_j) = \beta_{i1} [E(R_m) - R_f] + \sum_{j=2}^k \beta_{ij} E(F_j)$

$$4. \text{ Single-Index Model: } R_i - R_f = \beta_i (R_m - R_f) + e_i$$

$$5. \text{ Factor weight associated with each factor} = \frac{\text{Total Security Risk}}{\text{Total Market Risk}}$$

$$6. E(R_p) = R_f + \beta_p [E(R_m) - R_f] = R_f + (\omega_1 \beta_1 + \omega_2 \beta_2) [E(R_m) - R_f]$$

$$7. \text{ Asset's Beta} = \frac{\text{Correlation between asset and market} \times \text{S.D. of Asset}}{\text{S.D. of Market}}$$

$$8. \text{ Portfolio Beta} = \beta_p = \frac{\sum_{i=1}^n \omega_i \beta_i}{\sum_{i=1}^n \omega_i} = 1$$

$$9. \text{ Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p}$$

$$10. \text{ Treynor Ratio} = \frac{R_p - R_f}{\beta_p}$$

$$11. M^2 = (R_p - R_f) \frac{\sigma_m}{\sigma_p} - (R_m - R_f)$$

$$12. \text{ Jansen's Alpha} = \alpha_p = R_p - [R_f + \beta_p (R_m - R_f)]$$

$$13. \text{ Security Characteristic Line (SCL)} = R_i - R_f = \alpha_i + \beta_i (R_m - R_f)$$

$$14. \text{ Weight of Non-market security should be proportional to } \frac{\text{Alpha of Security } i}{\text{Nonsystematic variance of Security } i} = \alpha_i / \sigma_i^2$$

$$15. \text{ Total Weight of Non-market security should be proportional to } = \frac{\sum_{i=1}^N \omega_i \alpha_i}{\sum_{i=1}^N \omega_i^2 \sigma_i^2}$$

$$16. \text{ Information Ratio} = \frac{\text{Alpha of Security } i}{\text{Nonsystematic Risk of Security } i}$$

$$17. \text{ Expected Return of Portfolio (under Arbitrage Pricing Model)} = E(R_p) = R_f + \lambda_1 \beta_{p,1} + \dots + \lambda_k \beta_{p,k}$$

$$18. \text{ Return on an Asset in excess of 1-Month T-Bill Return (under four factor model)} = E(R_{it}) = \alpha_i + \beta_{i,\text{MKT}} \text{MKT}_t + \beta_{i,\text{SMB}} \text{SMB}_t + \beta_{i,\text{HML}} \text{HML}_t + \beta_{i,\text{UMD}} \text{UMD}_t$$

Reading 45: Basics of Portfolio Planning & Construction

1. Investor's Expected Utility from Portfolio = $U_p = E(R_p) - \lambda \sigma_p^2$
2. Tactical Asset Allocation (TAA) Return contribution = Actual return of the portfolio – Return that would have been earned if the asset class weights were equal to the policy weights

Reading 46: Market Organization & Structure

1. Total return to a Leveraged Stock Purchase = $\frac{\text{Remaining Equity} - \text{IO}}{\text{IO}}$ where,
Remaining Equity = IO – Purchase commission + (-) Trading g(l) – Margin i paid + Div received – Sales commission paid OR

Remaining Equity = Proceeds on sale –
Payoff loan – Margin i paid + Div received
– Sales commission paid

2. ROE (based on leverage alone)
= Leverage (in times) $\times$ stock price return
(in %)

3. Price of stock below which a margin call
will take place (P):

$$\frac{\text{Initial margin (\$)} + (P - \text{Initial Stock Price})}{P} =$$

Maintenance Margin Requirement (%)

4. Total cost of placement to the issuing firm
in IPO (\$) = Gross proceeds received by the issuing
firm – Net proceeds received by the issuing
firm

5. Total cost of placement to the issuing firm
in IPO (%) = $\frac{\text{Gross proceeds received by IF} - \text{Net proceeds received by IF}}{\text{Net proceeds received by IF}}$

where IF = Issuing firm

6. Max leverage ratio = $\frac{100\%}{\% \text{ of Equity}}$

7. Max leverage ratio for position financed by
min margin requirement = $\frac{1}{\text{Min margin requirement}}$

Reading 47: Security Market Indices

1. Value of a price return index =

$$V_{PRI} = \frac{\sum_{i=1}^N n_i P_i}{D}$$

For Single Period:

2. % Change in value of Price return of

$$\text{index Portfolio} = PR_I = \frac{V_{PRI1} - V_{PRI0}}{V_{PRI0}}$$

3. Price Return (Ind constituent security): PR_I

$$= \frac{P_{i1} - P_{i0}}{P_{i0}}$$

4. Price return of the index: $PR_I =$

$$\sum_{i=1}^N w_i \left(\frac{P_{i1} - P_{i0}}{P_{i0}} \right)$$

5. % Δ in value of Total return of Index

$$\frac{V_{PRI1} - V_{PRI0} + Inc_I}{V_{PRI0}}$$

6. Total return of each security = $TR_i =$

$$\frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}}$$

$$\text{Total Return} = \sum_{i=1}^N w_i \left(\frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}} \right)$$

Over Multiple Time Periods:

7. Value of Price Return index at time $t =$
 $V_{PRIT} = V_{PRI0} (1 + PR_{I1}) (1 + PR_{I2}) \dots (1 + PR_{IT})$

8. Value of Total Return index at time $t =$
 $V_{TRIT} = V_{TRI0} (1 + TR_{I1}) (1 + TR_{I2}) \dots (1 + TR_{IT})$

9. Weight of security i under price weighting
 $= \frac{\text{Price of security } i}{\text{Sum of all prices of constituent securities}}$

10. Weight of security i under equal weighting
 $= \frac{1}{\text{No of securities in the index}}$

11. Weight of security i under market-cap
weighting = $\frac{\text{No of shares o/s of Si} \times \text{Share price of Si}}{\sum_{i=1}^N (\text{No of shares o/s of Si} \times \text{Share price of Si})}$
Where Si = Security i

12. Weight of Si under Mkt Cap weighting = $\frac{\text{Fraction of shares o/s mkt float} \times \text{of shares Si} \times \text{Share price of security } i}{\sum (\text{Fraction of shares o/s Mkt float} \times \text{of shares o/s of Si} \times \text{Share price of security } i)}$

13. Fundamental weight on security $i =$
 $\frac{\text{Fundamental size measure of company } i^*}{\sum_{i=1}^N (\text{Fundamental size measure of company } i)}$

*Book value, cash flow, revenues, earnings, dividends, & number of employees.

Reading 48: Market Efficiency

Reading 49: Overview of equity Securities

- Equity security's Total Return =
$$\frac{\text{Sale P of a share} - \text{Purchase P of a share} + \text{cash/stock Div}}{\text{Purchase price of a share}}$$
- ROE in yr t =
$$\frac{\text{NI (for Ordinary Shareholders) in yr t}}{\text{Avg Total BV of Equity}}$$

OR
$$\text{ROE} = \frac{\text{NI (for Ordinary Shareholders) in yr t}}{\text{Shareholders' equity at beg of yr t}}$$
- MV of equity = Mkt price per share $\times$ Shares O/s
- BV of equity per share =
$$\frac{\text{Total SH' equity}}{\text{Shares o/s}}$$
- Price-to-book ratio =
$$\frac{\text{Market price per share}}{\text{BV of equity per share}}$$
- ROE = Net profit margin $\times$ Asset turnover $\times$ Financial leverage =
$$\left(\frac{\text{Net earnings}}{\text{Net sales}} \right) \times \left(\frac{\text{Net sales}}{\text{Avg total assets}} \right) \times \left(\frac{\text{Avg total assets}}{\text{Avg common equity}} \right)$$

Reading 50: Introduction to Industry & Company Analysis

Reading 51: Equity Valuation: Concepts & Basic Tools

- Value of a share of stock today =
$$\sum_{t=1}^{\infty} \frac{\text{Expected dividend in yr } t}{(1 + \text{required ROR on stock})^t}$$

If an investor intends to buy and hold a share for 1 yr:

- Value of a share of stock today =
$$\frac{\text{Expected Div in 1 yr} + \text{Expected selling price in 1 year}}{(1 + \text{req RoR on stock})^1}$$
- Value of a share of stock for n holding period or investment horizon =
$$\sum_{t=1}^n \frac{\text{Expected Div in yr } t}{(1 + \text{req R on stock})^t} + \frac{\text{Expected price in n periods}}{(1 + \text{req R on stock})^n}$$
- CFO = NI + Non-cash exp – Invst in WC
- FCFE = CFO – FCInv + Net Borrowing
- Value of a share for a non-div-paying stock =
$$\sum_{t=1}^{\infty} \frac{\text{FCFE in year } t}{(1 + \text{req R on stock})^t}$$
- Req RoR on share_i = Current expected R_f rate + Beta_i [MRP]

- Value of a pref stock (non-callable, non-convertible) =

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{D_0(1+0)}{r-0} = \frac{D_0}{r}$$

- Value of a pref stock (non-callable, non-convertible) with maturity at time n =

$$V_0 = \sum_{t=1}^n \frac{D_0}{(1+r)^t} + \frac{F}{(1+r)^n}$$

Gordon Growth Model:

- Value of a share of stock =

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}, \quad g < r$$

- Sustainable dividend growth rate = $g = \text{ROE} \times b$
where $b = \text{earnings retention rate} = (1 - \text{Dividend payout ratio})$

Two-stage valuation model:

- Value of share today = $V_0 =$

$$V_0 = \sum_{i=1}^n \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

$$V_n = \frac{D_{n+1}}{r - g_L}$$

$$D_{n+1} = D_0(1+g_s)^n(1+g_L)$$

- Justified P/E = $\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{p}{r-g}$

- EV = MV of stock + MV of debt – Cash and cash Equivalents

- Asset-based value = Value of Equipment and inventory – Value of Liabilities

Reading 52: Fixed Income Securities: Defining Elements

- Inf adj Principal amount of a zero-coupon-indexed bond = $[\text{Par value} \times (1 + \text{CPI})]$
- Inf adj coupon payment for an interest-indexed bond = $[(\text{coupon rate} \times \text{Par value}) \times (1 + \text{CPI})]$

3. Inf adj Principal amount of a capital-indexed bond
 $= [\text{Par value} \times (1 + \text{CPI})]$
4. Inflation adjusted coupon payment for a capital-indexed bond
 $= [\text{Par value} \times (1 + \text{CPI})] \times \text{coupon rate}$

Reading 53: Fixed Income Markets: Issuance, Trading & Funding

Reading 54: Introduction to Fixed Income Valuation

1. Amount of discount below par value = Present value of deficiency
2. Present value of deficiency =

$$\sum_{t=1}^n \frac{(\text{Coupon rate} - \text{Market discount rate}) \times \text{Par value}}{(1 + \text{Market discount rate})^t}$$
3. Bond price =

$$PV = \frac{PMT}{(1+r)^1} + \frac{PMT}{(1+r)^2} + \dots + \frac{PMT + FV}{(1+r)^N}$$
4. % Price change = $\frac{\text{New price} - \text{Old price}}{\text{Old price}}$
5. Bond price (given sequence of spot rates)
 $= PV =$

$$\frac{PMT}{(1+Z_1)^1} + \frac{PMT}{(1+Z_2)^2} + \dots + \frac{PMT + FV}{(1+Z_N)^N}$$

6. Full price of bond = Flat price of bond + Accrued interest
7. Accrued interest = $AI = \frac{t}{T} \times PMT$
8. Full price of a fixed-rate bond between coupon payments = PV^{Full}

$$= \frac{PMT}{(1+r)^{1-t/T}} + \frac{PMT}{(1+r)^{2-t/T}} + \dots + \frac{PMT + FV}{(1+r)^{N-t/T}}$$
9. Full price of a fixed-rate bond between coupon payments
 $PV \times (1+r)^{t/T}$
10. Interpolated yield (say for 3-year, given market discount rates for 2 and 5 yrs) =
 (Average yield for 2 year bonds) + $\left(\frac{3-2}{5-2}\right) \times$
 (average yield for 5 year bonds – average yield for 2 year bonds)
11. $\left(1 + \frac{APR_m}{m}\right)^m = \left(1 + \frac{APR_n}{n}\right)^n$
12. Current yield =

$$\frac{\text{Sum of coupon payments received over the year}}{\text{Flat price}}$$
13. Price of Floating-rate note = $PV =$

$$\frac{\frac{(I+Qm) \times FV}{m}}{\left(1 + \frac{I+DM}{m}\right)^1} + \frac{\frac{(I+Qm) \times FV}{m}}{\left(1 + \frac{I+DM}{m}\right)^2} + \dots + \frac{\frac{(I+Qm) \times FV}{m} + FV}{\left(1 + \frac{I+DM}{m}\right)^N}$$

14. Price of Money Market Instrument =

$$PV = FV \times \left(1 - \frac{\text{Days}}{\text{Year}} \times DR\right)$$

15. Market Discount Rate =

$$DR = \left(\text{Year} / \text{Days}\right) \times \left(\frac{FV - PV}{FV}\right)$$

16. Price of Money Market Instrument =

$$PV = \frac{FV}{\left(1 + \frac{\text{Days}}{\text{Yr}} \times AOR\right)}$$

17. Add-on rate =

$$AOR = \left(\frac{\text{Yr}}{\text{Days}}\right) \times \left(\frac{FV - PV}{PV}\right)$$

Relation b/w two spot rates and Implied Forward Rate:

$$18. (1 + z_A)^A \times (1 + \text{IFR}_{A,B-A})^{B-A} = (1 + z_B)^B$$

Z-spread over the benchmark spot curve:

Price of a bond =

$$PV = \frac{PMT}{(1+z_1+Z)^1} + \frac{PMT}{(1+z_2+Z)^2} + \dots + \frac{PMT + FV}{(1+z_N+Z)^N}$$

19. OAS = Z-spread – Option value (bps per year)
20. G-spread = Yield-to-maturity on Corporate bond – Yield-to-maturity on a government bond
21. Interpolated Spread = I-spread = YTM of the bond - Linearly interpolated yield to the same maturity on an appropriate reference curve

Reading 55: Introduction to Asset Backed Securities

1. Loan-to-value ratio (LTV) = $\frac{\text{Amount of Mortgage}}{\text{Property Value}}$
2. Monthly CF for a MPS = Monthly CF of underlying pool of mortgages - Servicing fee - Other fees
3. Pass-through rate = Mortgage rate on the underlying pool of mortgages – Servicing Fee - Other fees
4. SMM = Pre-pmt for month ÷ (Beg mortgage balance for month – Scheduled principal re-pmt for month)
5. $\text{CPR} = 1 - (1 - \text{SMM})^{12}$
6. CF Construction (Monthly CF for MPS):

- Net interest = (Beg mortgage balance × Pass-through rate) / 12
- Scheduled principal re-pmt = Mortgage pmt – Gross *i*- pmt
- Gross *i*- pmt = (Beg mortgage balance × WAC) / 12
- Pre-pmt for month = SMM × (Beg mortgage balance for month – Scheduled principal re-pmt for month)
- Total principal re-pmt = Scheduled principal re-pmt + Prepayment
- Beg mortgage balance for the following month = Beg mortgage balance for the month – Total Principal Pmt
- Projected CF for MPS = Net *i*- pmt + Total principal re-pmt

$$7. \text{ DSC ratio} = \frac{\text{Property's annual NOI}}{\text{Debt service}}$$

Reading 56: Understanding Fixed Income Risk & Return

1. Interest-on-interest gain from compounding = Future value of reinvested coupons - Total amount of coupon payments
Where,
FV of Reinvested Coupons = $[\text{CR} \times (1 + \text{RR})^{n-1}] + [\text{CR} \times (1 + \text{RR})^{n-2}] + \dots + [\text{CR} \times (1 + \text{RR})^{n-n}]$

$$\text{Total Amount of Coupon Pmt} = \text{CR} \times \text{Par value} \times \text{No of periods}$$

RR = Re-investmnt rate per period

CR = coupon rate

2. Realized RoR on Bond =

$$\left[\frac{\text{Sum of Reinvested Coupons} + \text{Redemption of Principal at Maturity}}{\text{Bond Price}} \right]^{\frac{1}{n}} - 1$$

3. Carrying value of bond (if bond purchased below par) = Purchase price + Amortized amount of Discount
4. Carrying value of a bond (if bond purchased above par) = Purchase price – Amortized amount of Premium
5. Amortized amount for 1st year = Bond Price after 1-yr - Initial bond price
6. Capital g / (l) = Sale price of Bond after n years – Carrying value of Bond after n years

7. Macaulay Duration =

$$\text{MacDur} = \left\{ (1-t/T) \left[\frac{\frac{\text{PMT}}{(1+r)^{1/T}}}{\text{PV}^{\text{Full}}} \right] + (2-t/T) \left[\frac{\frac{\text{PMT}}{(1+r)^{2/T}}}{\text{PV}^{\text{Full}}} \right] + \dots + (N-t/T) \left[\frac{\frac{\text{PMT} + \text{FV}}{(1+r)^{N/T}}}{\text{PV}^{\text{Full}}} \right] \right\}$$

OR

$$\text{MacDur} = \left\{ \frac{1+r}{r} - \frac{1+r + [N \times (c-r)]}{c \times [(1+r)^N - 1] + r} \right\} - (t/T)$$

$$8. \text{ Modified D} = \frac{\text{MacDur}}{1+r}$$

$$9. \text{ Annualized Modified D} = \frac{\text{Modified Duration}}{\text{Periodicity of payment in a year}}$$

$$10. \% \Delta PV^{\text{Full}} = -\text{AnnModDur} \times \Delta \text{Yield}$$

$$11. \text{ Approx Modified D} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta \text{Yield}) \times (PV_0)}$$

$$12. \text{ Approx Mac Dur} = \text{Approx Mod Dur} \times (1 + r)$$

$$13. \text{ Effective D} = \frac{(PV_-) - (PV_+)}{2 \times (\Delta \text{Curve}) \times (PV_0)}$$

$$14. \text{ Macaulay D for a Zero-coupon bond} = \frac{N-t}{T}$$

$$15. \text{ Macaulay D for a Perpetual bond} = (1+r)/r$$

$$16. \text{ Avg Mod D for the Portf} = \left(\text{Mod D of Bond 1} \times \frac{\text{MV of Bond 1}}{\text{Total MV of Portf}} \right) + \left(\text{Mod D of Bond 2} \times \frac{\text{MV of Bond 2}}{\text{Total MV of Portf}} \right) + \dots + \left(\text{Mod D of Bond N} \times \frac{\text{MV of Bond N}}{\text{Total MV of Portf}} \right)$$

$$17. \text{ Money D} = \text{Annualized Mod D} \times \text{Full Bond Price}$$

$$18. \Delta \text{ Full price of Bond (in currency units)} \approx -\text{Money D} \times \Delta \text{ in annual YTM}$$

$$19. \text{ PVBP} = \frac{(PV_-) - (PV_+)}{2}$$

$$20. \text{ Basis Point Value (BPV)} = \text{Money duration} \times 0.0001 \text{ (1 bp)}$$

$$21. \text{ Bloomberg's Risk Statistic} = \text{PVBP} \times 100$$

$$22. \% \Delta PV^{\text{Full}} = (-\text{AnnModDur} \times \Delta \text{Yield}) + \left[\frac{1}{2} \times \text{AnnConvexity} \times (\Delta \text{Yield})^2 \right]$$

Or

$$\% \Delta PV^{\text{Full}} = (-\text{AnnModDur} \times \Delta \text{Yield}) + \left[\frac{1}{2} \times \text{AnnConvexity} \times (\Delta \text{Yield})^2 \right]$$

$$23. \text{ Approx. Convexity Adjustment} = \frac{(PV_-) + (PV_+) - [2 \times (PV_0)]}{(\Delta \text{Yield})^2 \times (PV_0)}$$

$$24. \text{ Convexity of a zero coupon bond} = \frac{[N - (t/T)] \times [N + 1 - (t/T)]}{(1+r)^2}$$

$$25. \text{ Money Convexity vs Money Duration} = \Delta PV^{\text{Full}} \approx -(\text{MoneyDur} \times \Delta \text{Yield}) + \left[\frac{1}{2} \times \text{MoneyCon} \times (\Delta \text{Yield})^2 \right]$$

$$26. \text{ Money Convexity of bond} = \text{Annual Convexity} \times \text{Full Price}$$

$$27. \text{ Effective Convexity} = \frac{[(PV_-) + (PV_+) - [2 \times (PV_0)]]}{(\Delta \text{Curve})^2 \times (PV_0)}$$

$$28. \text{ Duration Gap} = \text{Bond's Macaulay Duration} - \text{Investment Horizon}$$

Reading 57: Fundamentals of Credit Analysis

$$1. \text{ Expected Loss} = \text{Default Probability} \times \text{Loss Severity given Default}$$

$$2. \text{ Operating Profit Margin} = \frac{\text{Operating Income}}{\text{Revenue}}$$

$$3. \text{ EBITDA} = \text{Operating Income} + \text{Dep} + \text{Amort}$$

$$4. \text{ FCF} = \text{CFO} - \text{Cap exp} - \text{Div}$$

$$5. \text{ Capital expenditures} = \text{Additions to P\&E} + \text{Additions to product rights \& intangibles} - \text{Proceeds of sale of P\&E}$$

$$6. \text{ Total debt} = \text{ST debt} + \text{Current portion of LT debt} + \text{LT debt}$$

$$7. \text{ Capital} = \text{Debt} + \text{Equity}$$

$$8. \text{ Yield on Corp Bond} = \text{Rf rate} + \text{Expected Inf rate} + \text{Maturity P} + \text{Liquidity P} + \text{Credit spread}$$

$$9. \text{ Yield spread} = \text{Liquidity P} + \text{Credit spread}$$

10. Return impact for smaller spread $\Delta \approx \% \Delta$ in price $\approx -\text{Modified Duration} \times \Delta \text{Spread}$

11. Return impact for larger spread $\Delta \approx \% \Delta$ in price $\approx -(\text{Modified D} \times \Delta \text{Spread}) + \frac{1}{2} \text{Convexity} \times (\Delta \text{Spread})^2$

12. Secured debt leverage = $\frac{\text{Total secured debt}}{\text{EBITDA}}$

13. Senior unsecured leverage = $\frac{\text{Secured debt} + \text{Senior unsecured debt}}{\text{EBITDA}}$

14. Total Leverage = $\frac{\text{Total debt}}{\text{EBITDA}}$

15. Net Leverage = $\frac{\text{Total debt} - \text{Cash}}{\text{EBITDA}}$

Reading 58: Derivatives Markets and Instruments

1. Value of the contract to the 'Long' at expiration = $S_T - F_0(T)$

2. Value of the contract to the 'Short' at expiration = $F_0(T) - S_T$

3. Margin % in stock market = $\frac{\text{MV of Stock} - \text{MV of Debt}}{\text{MV of Stock}}$

4. Margin Call:

- Long position: Price $\downarrow$ that would trigger a margin call = IM req – MM req

- Short position: Price $\uparrow$ that would trigger a margin call = IM req – MM req

5. TED spread = LIBOR – T-Bill rate

6. At expiration (for option Buyer):

- Value of Call option = $C_T = \text{Max}(0, S_T - X)$
- Profit from Call option = $\text{Max}(0, S_T - X) - C_0$
- Value of Put option = $P_0 = \text{Max}(0, X - S_T)$
- Profit from Put option = $\text{Max}(0, X - S_T) - P_0$

7. At expiration (for option Seller):

- Profit from Call option = $-\text{Max}(0, S_T - X) + C_0$
- Profit from Put option = $-\text{Max}(0, X - S_T) + P_0$

8. To eliminate arbitrage opportunity:
Forward Price should be = Spot Price $\times (1 + \text{interest rate \%})^t$

Reading 59: Basics of Derivative Pricing & Valuation

1. Pricing of risky assets = $S_0 = \frac{E(ST)}{(1+r+\lambda)^T}$

2. Commodity = $F_{0,T} = S_0 e^{(r-\delta)T}$
where, δ = Convenience yield – Cost of carry

3. $S_0 = \frac{E(ST)}{(1+r+\lambda)^T} - \theta + \gamma$

where, θ (theta) = PV of the costs and γ (gamma) = PV of benefits

4. Arbitrage and Derivatives = Underlying asset + Opposite position in derivative = Underlying payoff – Derivative payoff = Rf return

5. Pricing and Valuation of Forward Contracts:

- At Expiration $F(0, T) = S_0 (1+r)^T$ or $S_0 = F(0, T) / (1+r)^T$
- Value of forward (long) during contract life (where $t < T$) = $V_t(0, T) = S_t - F(0, T) / (1+r)^{(T-t)}$
- Value of forward (short) during contract life (where $t < T$) = $V_t(0, T) = F(0, T) / (1+r)^{(T-t)} - S_t$
- Value of forward (long) at expiration (where $t = T$) = $V_T(0, T) = S_T - F(0, T)$
- Value of forward (long) at initiation (where $t = 0$) = $V_t(0, T) = S_0 - F(0, T) / (1+r)^T = 0$
- Forward price of an asset with benefits and/or costs = $(S_0 - \gamma + \theta) (1+r)^T = S_0 (1+r)^T - (\gamma - \theta) (1+r)^T$
- Value of Forward contract with benefits and/or costs during the life of the contract = $S_t - (\gamma - \theta) (1+r)^T - F(T) / (1+r)^{(T-t)}$

6. FRAs: An example of 3×9 FRA (read as three by nine):
- Contract expires in 90 days
 - Underlying loan settled in 270 days
 - Underlying rate is 180-day LIBOR
 - For Synthetic FRA (take long position in a 300-day Euro\$ T.D and short position in a 30-day Euro\$ T.D
 - For synthetic forward position in a 90-day zero-coupon that begins in 30 day (buy 120 day & sell 30 day (zero coupon bonds)
7. Pricing and Valuation of Swap Contract (a fixed for floating swap contract):
- Fixed Periodic rate =
- $$R_N = \frac{1 - Z_N}{Z_1 + Z_2 + \dots + Z_N}$$
- Where Z_n are n period zero coupon bonds (i.e. \$1 discount factors)
- $$Z_n = \frac{1}{1 + (L_n \times \text{days} / 360)}$$
- Value of a fixed rate side (per \$1 NP)
- $$= V_{\text{fixed rate}} = [\text{Fixed payment} \times (Z_1 + Z_2 + \dots + Z_N)] + (\$1 \times Z_N)$$
- Value of a floating rate side (per \$1 NP) = $V_{\text{floating rate}} = (\$1 + 1^{\text{st}} \text{ floating pmt}) \times Z_1$

Pricing and valuation of Options:

8. Payoff of Call options:

- At expiration call option = $c_T = \text{Max}(0, S_T - X)$
- Profit (call buyer) = $\text{Max}(0, S_T - X) - c_0$
- Profit (call seller) = $-\text{Max}(0, S_T - X) + c_0$

9. Payoff of Put options:

- $p_T = \text{Max}(0, X - S_T)$
- Profit (put buyer) = $\text{Max}(0, X - S_T) - p_0$
- Profit (put seller) = $-\text{Max}(0, X - S_T) + p_0$

10. Max Profit/Loss for Option writer/holder:

- Max profit of option seller/writer → Option premium.
- Max loss of option seller/writer → unlimited.
- Max loss of option holder → Option premium

Put-Call Parity

11. Protective Put

- Value PP = $p_0 + S_0$
- Payoff at expiration (put out-of-the-money) = S_T .
- Payoff at expiration (put in-the-money) = $(X - S_T) + S_T = X$.

12. Fiduciary Call

- Value FC = $c_0 + X / (1+r)^T$
- Payoff at expiration (when call out-of-the-money) = X .
- Payoff at expiration (call in-the-money) = $X + (S_T - X) = S_T$.

13. Put-Call Parity (to avoid arbitrage) = $c_0 + X / (1+r)^T = p_0 + S_0$

- Synthetic long position in a call =

$$C = p_0 + S_0 - \frac{X}{(1+r)^T}$$

- Synthetic long position in a put =

$$p_0 = c_0 - S_0 + \frac{X}{(1+r)^T}$$

- Synthetic long position in an

$$\text{underlying} = S_0 = c_0 + \frac{X}{(1+r)^T} - p_0$$

- Synthetic long position in a riskless

$$\text{bond} = \frac{X}{(1+r)^T} = p_0 + S_0 - c_0$$

14. Put-Call-Forward Parity = $F_0(T) / (1+r)^T + p_0 = c_0 + X / (1+r)^T$

15. Valuing a callable bond using Binomial Model:

- $R_u = R_d \times e^{2\sigma\sqrt{t}}$
- Value at time 0 = $V_0 = hS_0 - c_0$
- Value at time 1 will either $V_1^+ = hS_1^+ - c_1^+$ or $V_1^- = hS_1^- - c_1^-$

- If the portfolio was hedged, then V^+ would equal V^- .

$$\begin{aligned} V_1^+ &= V_1^- \\ \Rightarrow hS_1^+ - c_1^+ &= hS_1^- - c_1^- \\ \Rightarrow h &= \frac{c_1^+ - c_1^-}{S_1^+ - S_1^-} \end{aligned}$$

- Value of the call =

$$c_0 = \frac{\pi c_1^+ + (1 - \pi)c_1^-}{1 + r}$$

where

$$\pi = \frac{1 + r - d}{u - d}$$

- Value of the put =

$$p_0 = \frac{\pi p_1^+ + (1 - \pi)p_1^-}{1 + r}$$

Reading 60: Risk Management Applications of Option Strategies

1. For Call Option Buyer

- $c_T = \max(0, S_T - X)$
- When $S_T \leq X \rightarrow c_T = 0$
- When $S_T > X \rightarrow c_T = S_T - X$
- Value at expiration = c_T
- Profit = $c_T - c_0$
- Maximum profit = $\infty \rightarrow$ no upper limit
- Maximum loss = c_0
- Breakeven = $S_T^* = X + c_0$

2. For Call Option Seller

- $c_T = \max(0, S_T - X)$
- When $S_T \leq X \rightarrow c_T = 0$
- When $S_T > X \rightarrow c_T = S_T - X$
- Value at expiration = $-c_T$
- Profit = $-c_T + c_0$
- Maximum profit = c_0
- Maximum loss = $\infty \rightarrow$ no upper limit
- Breakeven = $S_T^* = X + c_0$

3. For Put Option Buyer

- $p_T = \max(0, X - S_T)$
- When $S_T < X \rightarrow p_T = X - S_T$
- When $S_T \geq X \rightarrow p_T = 0$
- Value at expiration = p_T
- Profit = $p_T - p_0$
- Maximum profit = $X - p_0$
- Maximum loss = p_0
- Breakeven = $S_T^* = X - p_0$

4. For Put Option Seller

- $p_T = \max(0, X - S_T)$
- When $S_T < X \rightarrow p_T = X - S_T$
- When $S_T \geq X \rightarrow p_T = 0$
- Value at expiration = $-p_T$
- Profit = $-p_T + p_0$
- Maximum profit = p_0
- Maximum loss = $X - p_0$
- Breakeven = $S_T^* = X - p_0$

5. Covered Call = Long stock position + Short call position

- Value at expiration = $V_T = S_T - \max(0, S_T - X)$
- When $S_T \leq X \rightarrow V_T = S_T$
- When $S_T > X \rightarrow V_T = S_T - S_T + X = X$
- Profit = $V_T - S_0 + c_0$
- Maximum Profit = $X - S_0 + c_0$
- Maximum Loss = $S_0 - c_0$
- Breakeven = $S_T^* = S_0 - c_0$

6. Protective Put = Long stock position + Long Put position

- Value at expiration: $V_T = S_T + \max(0, X - S_T)$
- When $S_T \leq X \rightarrow V_T = S_T + X - S_T = X$
- When $S_T > X \rightarrow V_T = S_T$
- Profit = $V_T - S_0 - p_0$
- Maximum Profit = ∞
- Maximum Loss = $S_0 + p_0 - X$
- Breakeven = $S_T^* = S_0 + p_0$

Reading 61: Introduction to Alternative Investments

- Total Return = Alpha R + Beta R
- Asset Based Valuation = Co value = Co's assets value - Co's liabilities value

Real Estate Valuation

3. Direct Cap Approach → Valuation of a property = $\frac{NOI}{\text{Capitalization Rate}}$ where
NOI = Gross potential income – Estimated vacancy losses – Estimated collective losses – Insurance – Property Taxes – Utilities – Repairs, maintenance exp.
4. Income Based Approach → FFO = NI + Dep exp on R.E + Def Tax charges – Gains from sales of R.E + losses from sale of R.E
5. AFFO = FFO – Recurring Cap exp
6. Asset based Approach → REIT's NAV = Estimated MV of REIT's total assets – Value of REIT's total liabilities.
7. Pricing of Commodity Futures Contracts:
Futures price $\approx$ Spot price $(1 + r) +$ Storage costs – Convenience yield
8. Roll yield = Spot price of a commodity – Futures contract price
or
Roll yield = Futures contract price with expiration date 'X' – Futures contract price with expiration date 'Y'.
9. Returns on a *passive* investment in commodity futures
= Return on the collateral + RP or convenience yield net of storage costs.
10. Sharpe ratio = $(\text{Investment return} - R_f \text{ return}) / \text{S.D. of return}$
11. Sortino Ratio = $(\text{Annualized RoR} - \text{Annualized } R_f \text{ rate}) / \text{Downside Deviation}$